

Chronic Vertigo - Medical Clinical Policy Bulletins

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Policy

Scope of Policy

This Clinical Policy Bulletin addresses chronic vertigo.

1. Medical Necessity

Aetna considers the following procedures as medically necessary (unless otherwise specified) for chronic vertigo:

1. Diagnosis and Evaluation of Chronic Vertigo or Ménière's Disease

1. Caloric vestibular testing
2. Dynamic or head shaking acuity testing
3. Head impulse or head thrust test
4. Optokinetic nystagmus test
5. Spontaneous nystagmus test
6. Vibration induced nystagmus testing (VIN), skull vibration induced nystagmus testing (SVINT), bone conduction vibration;

2. Maneuvers for Benign Paroxysmal Positioning Vertigo (BPPV)

1. Positional nystagmus test (Barany or Dix-Hallpike maneuver) for the diagnosis of BPPV
2. Use of the Epley maneuver (also known as canalith repositioning procedure) or the Semont maneuver for the treatment of BPPV when *both* of the following selection criteria are satisfied:
 1. Diagnosis of BPPV has been confirmed by a positive Hallpike test, *and*
 2. Member had symptoms of BPPV for at least 4 months.

The Epley maneuver and the Semont maneuver have not been demonstrated to be effective in persons with disorders of the central nervous system such as temporal lobe epilepsy, multiple sclerosis, cerebrovascular disease, vertiginous migraine, cerebellopontine angle

tumors, and primary or metastatic cerebellar lesions, based on neurological examination, magnetic resonance imaging, or history. For individuals with these diagnoses and all other indications, use of the Epley maneuver or the Semont maneuver is considered experimental, investigational, or unproven.

3. Vestibular Rehabilitation

For chronic vertigo / persistent postural perceptual dizziness (PPPD) when *all* of the following criteria are met:

1. Symptoms (e.g., vertigo and imbalance) have existed for more than 6 months; *and*
2. The member has confirmed diagnosis of a vestibular disorder or has undergone ablative vestibular surgery; *and*
3. The member has failed medical management (e.g., use of vestibular suppressant medications to reduce symptoms).

Vestibular rehabilitation is considered experimental, investigational, or unproven for all other indications because its effectiveness for indications other than the one listed above has not been established.

Note: Up to 12 visits (generally given 2 times a week for 6 weeks) are considered medically necessary initially. Up to 12 additional visits are considered medically necessary if, upon medical review, there is evidence of clinically significant improvement. If there is no evidence of improvement after 12 visits, additional visits are not considered medically necessary.

4. Vestibular Evoked Myogenic Potentials

Ocular and cervical vestibular evoked myogenic potentials (cVEMP and oVEMP) to evaluate persons with vertigo for superior semicircular canal dehiscence syndrome (SCDS) who have had a comprehensive evaluation (history, physical, audiometry, electro- or videonystagmography, electrocochleography, brainstem audiometry) and the results are inconclusive. See also [CPB 0181 - Evoked Potential Studies](#).

5. Electronystagmography (ENG) and Videonystagmography (VNG)

1. ENG for evaluation of persons with symptoms of vestibular disorders (dizziness, vertigo, disequilibrium or imbalance);
2. VNG as a medically necessary alternative to ENG for assessment of vestibular disorders.

ENG and VNG are considered experimental, investigational, or unproven for all other indications because their effectiveness for indications other than the ones listed above has not been established.

2. Experimental, Investigational, or Unproven

The following procedures are considered experimental, investigational, or unproven because the effectiveness of these approaches has not been established (not an all-inclusive list):

1. Balanceback Intuitive VNG device for the management of balance disorders
2. Balance Tracking System (BTrackS) for evaluation of risk of falling
3. Brainstem auditory evoked potentials (BAEPs) for evaluation of individuals with vertigo
4. Cochlear hydrops analysis masking procedure (CHAMP) testing in the evaluation of Ménière's disease
5. DizzyFix device for the treatment of BPPV
6. Dynamic posturography (including evaluation of risk of falling, and work-up of children with Chiari malformation type 1)
1. Dynamic posturography (also known as balance board testing, computerized dynamic posturography [CDP], equilibrium platform testing [EPT], and moving platform posturography) for the diagnosis and staging of members with Ménière's disease and other

balance disorders, for the differential diagnosis of multiple sclerosis and disequilibrium, and all other indications;

2. Sensory organization test (SOT), also known as the gans sensory organization performance test (SOP); modified clinical test of sensory interaction on balance (mCTSIB); and movement coordination test (MCT) are components of dynamic posturography, and are considered experimental, investigational, or unproven;
3. Biodex BioSway Balance System for balance assessment
7. Epley Omniax Repositioning Chair for the treatment of benign paroxysmal positional vertigo
8. GyroStim machine for the treatment of balance disorders and vestibular dysfunction
9. Mastoid oscillation (mastoid vibration) for persons treated with canalith repositioning procedure
10. Meniett low-pressure pulse generator for the treatment of Ménière's disease, nausea/vomiting, and tinnitus
11. Posterior semicircular canal occlusion for the treatment of BPPV.

3. Related Policies

- [CPB 0181 - Evoked Potential Studies](#)
 - [CPB 0299 - Tilt Table Testing](#)
 - [CPB 0406 - Tinnitus Treatments](#)
 - [CPB 0467 - Vestibular Autorotation Test \(VAT\)](#)
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CPT Codes / HCPCS Codes / ICD-10 Codes

Code	Code Description
<i>Maneuvers for Benign Paroxysmal Positioning Vertigo:</i>	
CPT codes covered if selection criteria are met:	
92532	Positional nystagmus test [Hallpike maneuver]
95992	Canalith repositioning procedure(s) (eg, Epley maneuver, Semont maneuver), per day
ICD-10 codes covered if selection criteria are met:	
H81.10 - H81.13	Benign paroxysmal vertigo
ICD-10 codes not covered for indications listed in the CPB (not all-inclusive):	
C71.6	Malignant neoplasm of cerebellum
C79.31 - C79.49	Secondary malignant neoplasm of brain and other and unspecified parts of nervous system [cerebellopontine angle tumor]

Code	Code Description
D21.0	Benign neoplasm of connective and other soft tissue of head, face, and neck [cerebellopontine angle tumor]
D32.0	Benign neoplasm of cerebral meninges [cerebellopontine angle tumor]
D33.3	Benign neoplasm of cranial nerves [cerebellopontine angle tumor]
D43.0 - D43.9	Neoplasm of uncertain behavior of brain and central nervous system [cerebellopontine angle tumor]
G35	Multiple sclerosis
G40.001 - G40.219	Epilepsy and recurrent seizures [temporal lobe epilepsy]
G43.001 - G43.919	Migraine [vertiginous]
I60.00 - I69.998	Cerebrovascular diseases
Vestibular Rehabilitation:	
CPT codes covered if selection criteria are met:	
92531	Spontaneous nystagmus, including gaze
92533	Caloric vestibular test, each irrigation (binaural, bithermal stimulation constitutes 4 tests)
92534	Optokinetic nystagmus test
92537	Caloric vestibular test with recording, bilateral; bithermal (ie, one warm and one cool irrigation in each ear for a total of four irrigations)
92538	monothermal (ie, one irrigation in each ear for a total of two irrigations)
92541	Spontaneous nystagmus test, including gaze and fixation nystagmus, with recording
92542	Positional nystagmus test, minimum of 4 positions, with recording
92544	Optokinetic nystagmus test, bidirectional, foveal or peripheral stimulation, with recording
92545	Oscillating tracking test, with recording
92546	Sinusoidal vertical axis rotational testing
+ 92547	Use of vertical electrodes (List separately in addition to code for primary procedure)
99173	Screening test of visual acuity, quantitative, bilateral [dynamic acuity testing]
Other CPT codes related to the CPB:	

Code	Code Description
97112	Therapeutic procedure, one or more areas, each 15 minutes; neuromuscular re-education of movement, balance, coordination, kinesthetic sense, posture, and/or proprioception for sitting and/or standing activities
HCPCS codes covered if selection criteria are met:	
S9476	Vestibular rehabilitation program, non-physician provider, per diem
ICD-10 codes covered if selection criteria are met:	
• F45.0 - - F45.1, • F45.8, • F45.9	Somatoform disorder [persistent postural perceptual dizziness (PPPD)]
H81.01 - H83.93	Disorders of vestibular function
I69.998	Other sequelae following unspecified cerebrovascular disease [chronic vertigo]
R26.0 - R26.9	Abnormalities of gait and mobility
R42	Dizziness and giddiness
Dynamic Posturography:	
CPT codes not covered for indications listed in the CPB:	
92548	Computerized dynamic posturography sensory organization test (CDP-SOT), 6 conditions (ie, eyes open, eyes closed, visual sway, platform sway, eyes closed platform sway, platform and visual sway), including interpretation and report;
92549	with motor control test (MCT) and adaptation test (ADT)
ICD-10 codes not covered for indications listed in the CPB (not all-inclusive):	
G35	Multiple sclerosis
G93.5	Compression of brain
H81.01 - H83.93	Disorders of vestibular function
I69.998	Other sequelae following unspecified cerebrovascular disease [chronic vertigo]
R26.0 - R26.9	Abnormalities of gait and mobility

Code	Code Description
R27.0 - R27.9	Other lack of coordination
R29.6	Repeated falls
R42	Dizziness and giddiness
Z91.81	History of falling
Sensory Organization Test (SOT):	
No specific code	
Meniett Low-Pressure Pulse Generator:	
HCPCS codes not covered for indications listed in the CPB:	
A4638	Replacement battery for patient-owned ear pulse generator, each
E2120	Pulse generator system for tympanic treatment of inner ear endolymphatic fluid
ICD-10 codes not covered for indications listed in the CPB (not all-inclusive):	
H81.01 - H81.09	Disorders of vestibular function
H93.11 - H93.19	Tinnitus
H93.A1 - H93.A9	Pulsatile tinnitus
R11.0 - R11.2	Nausea and vomiting
Videonystagmography and Electronystagmography:	
No specific code	
CPT codes covered if selection criteria are met:	
92541 - 92546	Vestibular function tests, with recording (eg, ENG)
+ 92547	Use of vertical electrodes (List separately in addition to code for primary procedure)
ICD-10 codes covered if selection criteria are met:	
H81.01 - H83.93	Vertiginous syndromes and other disorders of vestibular system

Code	Code Description
I69.998	Other sequelae following unspecified cerebrovascular disease [chronic vertigo]
R42	Dizziness and giddiness
<i>Cochlear Hydrops analysis masking procedure (CHAMP):</i>	
No specific code	
ICD-10 codes not covered for indications listed in the CPB (not all-inclusive):	
H81.01 - H81.09	Meniere's disease
<i>Brainstem Auditory Evoked Potentials (BAEPs):</i>	
CPT codes not covered for indications listed in the CPB:	
92650	Auditory evoked potentials; screening of auditory potential with broadband stimuli, automated analysis
92651	Auditory evoked potentials; for hearing status determination, broadband stimuli, with interpretation and report
92652	Auditory evoked potentials; for threshold estimation at multiple frequencies, with interpretation and report
92653	Auditory evoked potentials; neurodiagnostic, with interpretation and report
ICD-10 codes not covered for indications listed in the CPB (not all-inclusive):	
H81.01 - H81.4	Vertigo
<i>Biodex BioSway Balance System:</i>	
CPT codes not covered for indications listed in the CPB:	
<i>Biodex BioSway Balance System - no specific code:</i>	
<i>Posterior semicircular canal occlusion:</i>	
CPT codes not covered for indications listed in the CPB:	

Code	Code Description
<i>Posterior semicircular canal occlusion - no specific code:</i>	
<i>Cervical and ocular vestibular evoked myogenic potentials:</i>	
CPT codes covered if selection criteria are met::	
92517	Vestibular evoked myogenic potential (VEMP) testing, with interpretation and report; cervical (cVEMP)
92518	Vestibular evoked myogenic potential (VEMP) testing, with interpretation and report; ocular (oVEMP)
92519	Vestibular evoked myogenic potential (VEMP) testing, with interpretation and report; cervical (cVEMP) and ocular (oVEMP)
ICD-10 codes covered if selection criteria are met:	
H83.8X1- H83.8X9	Other specified diseases of inner ear [semicircular canal dehiscence syndrome (SCDS)]
R42	Dizziness and giddiness
<i>Balanceback intuitive VNG device:</i>	
CPT codes not covered for indications listed in the CPB:	
<i>Balanceback intuitive VNG device -no specific codes</i>	
ICD-10 codes not covered for indications listed in the CPB:	
R26.0 - R26.9	Abnormalities of gait and mobility
<i>Balance tracking system (Btracks):</i>	
CPT codes not covered for indications listed in the CPB:	
<i>Balance tracking system (Btracks) - no specific codes</i>	
ICD-10 codes not covered for indications listed in the CPB:	
R29.6	Repeated falls
Z91.81	History of falling

Code	Code Description
<i>Epley omni-ax repositioning chair:</i>	
CPT codes not covered for indications listed in the CPB:	
<i>Epley omni-ax repositioning chair -no specific codes</i>	
ICD-10 codes not covered for indications listed in the CPB:	
H81.10 – H81.13	Benign paroxysmal vertigo
<i>GyroStim machine:</i>	
HCPCS codes not covered for indications listed in the CPB:	
<i>GyroStim machine - No specific code</i>	
ICD-10 codes not covered for indications listed in the CPB:	
H81.01 – H81.93	Disorders of vestibular function
R26.0 – R26.9	Abnormalities of gait and mobility

Background

Vertigo can be described as a dizzy or spinning sensation. Some people perceive self-motion whereas others perceive motion of the environment. Individuals may experience vertigo as an illusion of motion, vague dizziness, imbalance, disorientation, transient spinning or a sense of swaying or tilting.

Vertigo may be caused by any number of conditions and is a symptom rather than a diagnosis. Once a diagnosis has been identified, treatment is focused on the specifics of the disease/disorder, relief of symptoms and promotion of recovery. The treatment also depends on whether the patient is suffering from acute or chronic symptoms. Acute vertigo will present as isolated spells and has a distinct beginning and end whereas chronic vertigo is continuous and/or recurring.

Vertigo is the predominant symptom of vestibular dysfunction and can be associated with health conditions such as, but not limited to, Ménière's disease and benign paroxysmal positional vertigo (BPPV).

Ménière's Disease is a disorder of the inner ear that may be associated with intermittent hearing loss, a sensation of ear fullness, pressure or pain, dizziness and/or a roaring sound in the ears called tinnitus. One or both ears may be affected.

Benign paroxysmal positioning vertigo (BPPV, also known as cupulolithiasis or benign paroxysmal positioning nystagmus) is believed to be a mechanical disorder of the inner ear as a consequence of degenerated material lodging in the posterior canal of the ear.

Vestibular Evaluation

The Hallpike maneuver is a specific clinical balance test that when positive, is diagnostic of BPPV. The classical nystagmus (an involuntary, rapid, rhythmic movement of the eyeball, which may be horizontal, vertical, rotatory, or mixed) occurs when the patient's head is rapidly reclined and turned to the affected side. The Semont maneuver and the Epley maneuver (also known as canalith repositioning procedure) are a series of head manipulations performed by trained physicians in an attempt to move the degenerated material along the posterior canal and out its opening, thus eliminating the symptoms.

Particle repositioning maneuvers (Canalith repositioning procedures) are used to manage episodes of BPPV. Canalith refers to collections of calcium in the inner ear. Particle repositioning maneuvers include, but are not limited to, the Brandt-Daroff exercises, the Epley maneuver and the Semont maneuver. The Brandt-Daroff Exercise is a positioning method for treating BPPV usually performed in the home; the exercises involve sitting up and lying down on a bed with the head in various positions to reposition particles in the ear. The Epley Maneuver is a particle repositioning maneuver that involves sequential movement of the head into four positions, staying in each position for approximately 30 seconds. The Semont Maneuver is a procedure in which the individual is rapidly moved from lying on one side to lying on the other side. The theory behind these therapies is that through a series of rotational movements, the particles will be cleared out of the semicircular canals via the common crus of the utricle where they will no longer have an impact on the dynamics of the semicircular canals. All of these maneuvers are generally well tolerated; however, they sometimes cause a migration of debris into the anterior and horizontal canals which cause other variants of positional vertigo.

There is sufficient evidence that the Hallpike maneuver is effective in diagnosing patients with BPPV. There is also enough scientific data to support the safety and effectiveness of the Semont maneuver and the Epley maneuver for the treatment of patients with this condition. Treatment usually requires a single session. Additional 1 to 2 sessions over a 2-week period may be necessary if the patient's condition does not improve or if the condition recurs after the initial session. Mastoid vibration should not be used in conjunction with the Epley maneuver in patients with perilymphatic fistula or a history of retinal detachment.

The American Academy of Neurology (AAN)'s guideline on therapies for BPPV (Fife et al, 2008) reported strong evidence supporting the canalith repositioning procedure (CRP) as a safe and effective treatment that should be offered to patients of all ages with posterior semicircular canal BPPV. Semont's maneuver is possibly effective. There was insufficient evidence to establish the relative efficacy of the Semont maneuver to CRP, nor was there enough evidence to recommend a specific maneuver for horizontal or anterior canal BPPV. The AAN guideline also noted that mastoid oscillation (i.e., the use of an oscillator placed on the mastoid process to enhance the effectiveness of CRP) is probably of no added benefit to patients treated with CRP.

The DizzyFix is a device designed to train patients to perform the particle repositioning maneuver (PRM), which helps to treat the most common cause of vertigo known as BPPV.

Bromwich et al (2008) developed and tested a completely new dynamic visual device for the home treatment of BPPV. These researchers designed and manufactured a new device (the DizzyFix) to assist in the performance of the PRM. A total of 50 healthy volunteers were taught the PRM, 50 % using the new device. At 1 week, these investigators compared the PRM performance between the device and non-device user groups. Main outcome measure was performance of the PRM as graded on an 11-point scale. DizzyFix users in phase I scored significantly higher on their PRM performance compared with controls ($p = 0.0001$). The authors concluded that the use of DizzyFix enables volunteers to conduct a correct PRM on their own. This is a significant improvement from written instructions or in-office training. This report appears to be a feasibility study in which healthy subjects were used. It did not provide clinical data regarding the effectiveness of the DizzyFix in treating patients BPPV.

Bromwich et al (2010) tested the effectiveness of the DizzyFix for the home treatment of BPPV. A total of 40 patients with active BPPV were included in this cohort study. Main outcome measure was the Dix-Hallpike maneuver at 1 week after treatment. Patients using the home treatment device had no evidence of nystagmus in post-treatment Dix-Hallpike maneuvers at 1 week in 88 % of cases. This rate was comparable to standard treatment. There were no significant complications. The authors concluded that the use of this device enables patients with an established diagnosis of posterior canal BPPV to safely conduct an effective PRM and achieve success rates similar to those found with the standard Epley maneuver. This was a small study with a short follow-up period; its

findings need to be validated by other investigators through well-designed studies.

The positional nystagmus test (Barany or Dix-Hallpike maneuver) involves moving the individual rapidly from the sitting to the lying position with the head tilted downward off the table at 45 degrees and rotated 45 degrees to one side to assess whether the eyes can maintain a static position when the head is in different position.

Silva et al (2011) discussed the current options available to manage BPPV. These investigators reviewed 2 recent guidelines regarding the evaluation and treatment of BPPV. The first one was published by the AAO-HNS and the other by the AAN. Only the AAO-HNS guidelines recommend the Dix-Hallpike test for the diagnosis of BPPV. Only canalith repositioning maneuver, Semont maneuver and vestibular rehabilitation had showed some benefit and were recommended as good treatment options.

Other methods for the evaluation of chronic vertigo and/or Ménière's disease may be generally referred to as vestibular evaluation and include, but may not be limited to:

Saccadic Testing

It is a general term used to describe various evaluations for nystagmus. A saccade is a small rapid jerky movement of the eye as it jumps from fixation on one point to another (as in reading). Nystagmus is uncontrollable movements of the eyes that may be side to side (horizontal), up and down (vertical) or rotary (torsional). Nystagmus is a type of saccade.

Caloric Vestibular Test

Each ear is separately irrigated with cold water and warm water, for a total of four irrigations, to create nystagmus in the individual. The patient is observed for any difference between the reaction of the right and the left sides.

Dynamic or Head Shaking Visual Acuity Test

By having the individual look at an eye chart in the distance wearing their customary distance vision eyeglasses, the individual reads the eye chart while their head is shaken continuously over a small range. Then, the individual reads the chart again while their head is still. A computerized system may be utilized as well to test for dynamic visual acuity.

Head Impulse or Head Thrust Test

By instructing the individual to keep their eyes on a distant target while wearing their usual prescription eyeglasses, the head is then turned quickly and unpredictably by the examiner. The normal response is that the eyes remain on the target.

Optokinetic Nystagmus Test

A rotating drum made of alternating light and dark vertical stripes is placed in front of the patient and the patient is instructed to stare at the drum without focusing on any one stripe. The eyes are observed for nystagmus while the drum is rotated in one direction. The direction of the drum is reversed. No electrodes are used.

Spontaneous Nystagmus Test

The individual's eyes are observed for spontaneous nystagmus as the individual is asked to look straight ahead, 30 degrees to 45 degrees to the right and 30 degrees to 45 degrees to the left. No electrodes are used and no recording made.

Vestibular Autorotation Test (VAT)

The individual wears a lightweight head strap with five electrodes which monitors head and eye movements. While following a moving target with the eyes, the individual moves the head back and forth or up and down in time with gradually accelerating computer generated tones.

Vestibular Evoked Myogenic Potential (VEMP)

Used to determine the function of the otolithic organs (utricle and saccule) of the inner ear. Headphones are placed over the ears and small electrodes are attached with an adhesive to the skin over the neck muscles. When sound is transmitted through the headphones, the electrodes record the response of the muscle to the vestibular stimuli.

Vibration Induced Nystagmus Testing (VIN), Skull Vibration Induced Nystagmus Testing (SVINT), Bone Conduction Vibration

A vibrating tuning fork or a battery operated vibrating device is placed at various points on the head and neck, usually the mastoid bone, while eye movements are observed.

Electronystagmography and Videonystagmography

Electronystagmography (ENG) is used to assess patients with vestibular disorders (e.g., dizziness, vertigo, or balance dysfunction). It provides objective testing of the oculomotor and vestibular systems. In general, the traditional ENG consists of the following 3 components:

- Caloric stimulation of the vestibular system; and
- Oculomotor evaluation (pursuit and saccades); and
- Positioning/positional testing.

With electronystagmography (ENG), eye movements are recorded and analyzed via small electrodes placed on the skin around the eyes. The testing is generally the same as the caloric vestibular test, optokinetic nystagmus test, positional nystagmus test and/or spontaneous nystagmus test; however in this variation, the results are recorded in addition to being observed.

Although ENG can not be used to ascertain the specific site of lesion, the information gathered can be integrated with clinical history, symptoms, and other test results to help in diagnosis. Comparing results obtained from various subtests of an ENG evaluation aids in determining if a disorder is central or peripheral. In peripheral vestibular disorders, the side of lesion can be inferred from the results of caloric stimulation and, to some degree, from positional findings. An ENG evaluation can also be useful in ruling out potential causes of dizziness.

While ENG is the most commonly used clinical test to evaluate vestibular function, normal ENG test results do not necessarily mean that a patient has typical vestibular function. Moreover, ENG abnormalities can be useful in the diagnosis and localization of site of lesion. However, many abnormalities are non-localizing; thus, the clinical history and otological examination of the patient are very important in formulating a diagnosis and treatment plan for a patient who presents with dizziness or vertigo.

Conventional ENG entails the use of electro-oculography to objectively record eye movements. This recording relies on the dipole of the eye (the corneal-retinal potential difference; the cornea is electro-positive relative to the retina). With a fixed recording site, voltage differences can be recorded for eye movements. Small electrodes are placed around the patient's eyes to record the corneal-retinal potential differences. By placing electrodes on both a horizontal and vertical axis around the eyes, tracings are produced for eye movements on both axes (Markley, 2007; Worden and Blevins, 2007; Shoup and Townsley, 2008).

Videonystagmography (VNG) is a technology for evaluating inner ear and central motor functions. Videonystagmography (VNG) is similar to electronystagmography, but eye movements are recorded by an infrared video camera mounted inside goggles that the patient wears instead of sticky-patch electrodes. The testing is generally the same as the caloric vestibular test, optokinetic nystagmus test, positional nystagmus test and/or spontaneous nystagmus test; however in this variation, the results are recorded in addition to being observed.

Ganança and colleagues (2010) compared literature information on the similarities, differences, advantages and disadvantages between VNG and ENG. These investigators noted that VNG and ENG are very helpful methods for evaluating balance disorders, due to their capacity to recognize signs of peripheral or central vestibular dysfunction and to

pinpoint the side of the lesion. Major advantages of VNG are related to calibration, temporo-spatial resolution, and recording of horizontal, vertical and torsional eye movements. The authors concluded that VNG is a new technology that presents advantages in the evaluation of eye movements; however, despite its disadvantages, ENG is still considered a valuable test in the clinical setting.

Dynamic Posturography

Dynamic posturography has been used for evaluation of suspected vestibular disorders. This diagnostic test employs a force platform and visual stimuli to measure the contributions to balance of vision, somatosensation, and vestibular sensation. The test measures postural stability (body sway), which is a functional indicator of balance.

Computerized dynamic posturography (CDP) or dynamic posturography is designed to help determine the severity of balance problems, estimate prognosis and plan treatment; it is also referred to as balance board testing, equilibrium platform testing or visual vertical testing. Dynamic posturography has been proposed for use in balance retraining and reassessment of treatment progress as well as to diagnose balance disorders caused by visual, vestibular or somatosensory (proprioceptor) problems. Dynamic posturography has also been proposed for use in the treatment of such problems.

Dynamic posturography is usually divided into 2 parts;

The former test alters proprioceptive and visual inputs, and determines the effects on equilibrium and on-feet anterior/posterior sway. The latter test assesses muscular reaction to various surface alterations induced by the equipment.

The protocol of the SOT is made up of 6 situations: Condition 1 allows the subject to stand on a flat, firm surface with eyes open, therefore, all sensory modalities are available for maintenance of balance. Condition 2 is identical to that of Condition 1 except that the subject's eyes are closed (No Visual Input). The first 2 conditions provide a baseline measure of the subject's stability. In Condition 3, the support surface is fixed and the visual surround is sway-referenced (Inaccurate Visual Input). This situation creates a visual conflict by moving the visual surround which the patient is watching as he/she moves. Thus, if one sways posteriorly by 3 degrees, the visual surround moves by the same magnitude in the same direction. This condition requires the subject to disregard the visual stimulus and utilize the proprioceptive and vestibular systems to control balance. Differences in stability observed during the first 3 conditions will reveal if the subject needs normal vision to maintain balance and suppress the influence of inaccurate visual cues. In Conditions 4, 5, and 6, the support surface is fixed (sway-referenced) while the visual condition is varied as in Conditions 1, 2, and 3. Thus, the visual surround is fixed and the patient receives accurate visual and vestibular inputs in Condition 4 (inaccurate proprioceptive input). Under Condition 5 (no visual input and inaccurate proprioceptive input), the subject's eyes are closed, thus the only information available is through the vestibular input. Under Condition 6 (inaccurate visual and proprioceptive inputs), the patient is left essentially with the vestibular system to achieve postural control. The key difference between Conditions 5 and 6 is that the subject has no visual cues in the former, whereas he/she has inaccurate visual cues in the latter. Because proprioceptive information is distorted in both situations, these 2 tests are designed to isolate the contributions from the vestibular system. In most patients with peripheral or central vestibular disorders, results from both Conditions 5 and 6 are abnormal, although not always of the same magnitude.

The patient is usually subjected to each of these 6 tests in separate 20-second trials, and each condition is repeated 3 times to ensure reliable outcomes and to determine adaptation with repeated testing. A separate equilibrium score is computed for each 20-second trial, with a score of 100 indicating no sway, and a score of 0 indicating the patient loses balance, namely, sway that exceeds the limits of stability (8.5 degrees anteriorly and 4 degrees posteriorly). In addition to equilibrium scores, COG alignment and the extent of hip versus ankle movement strategy are also recorded for each trial. Center of gravity alignment is represented in degrees of offset from the centered position. A low strategy score of near 0 represents a predominance of movement about the hip, while a high score of near 100 represents a predominance of movement about the ankle. Results of each condition are judged normal or abnormal by comparing the patient's average score with those obtained from age-matched normal subjects. Normal limits for a given age group are those attained or exceeded by 95% of individuals.

The modified clinical test of sensory interaction on balance (mCTSIB) is a simplified derivative of the SOT. Although the mCTSIB data set can document the presence of sensory dysfunction, it cannot provide impairment information specific to an individual sensory system. The information provided is designed to

A modification of the original CTSIB or "Foam and Dome", the mCTSIB eliminates the "dome" and adds computerized analysis of the patient's functional balance control to quantify postural sway velocity during the 4 sensory conditions:

The second part of the DP evaluation is the MCT, which examines coordination of lower limbs under various perturbations that create anterior or posterior sway, thus, necessitating a recovery response from the subject. It consists of a series of sudden forward and backward jerks of the platform. The perturbations are presented at 3 intensities in sets of 3 trials. Amplitudes of these sudden movements of the platform are height normalized to provide small perturbation (0.7 degree sway), medium perturbation (1.8 degrees sway), and large perturbation (3.2 degrees sway). In general, the duration of these 3 perturbations are 250, 300, and 400 msec, respectively. Results are analyzed in terms of latency, amplitude, and symmetry of motor responses. Latency (in msec) is a measure of the time interval from the commencement of perturbation to the moment when the subject begins to actively resist the induced sway - forward sway for backward jerk and backward sway for forward jerk. It is attained by averaging the performance of the left and right feet. Amplitude measures the muscle strength of responses to the induced forward or backward sway. Symmetry compares the strength of active forces generated by each leg against the force-plate. In normal individuals, response strength varies within 25 % of being identical in both perturbation directions and for all perturbation intensities.

Dynamic posturography is an evolving technology and there is insufficient peer-reviewed medical literature that addresses its clinical usefulness. CMS's Technology Advisory Committee recently concluded that there is insufficient evidence supporting computerized dynamic posturography's effectiveness for diagnosing balance disorders, or for predicting or influencing the prognosis. Prospective studies are needed to establish the role of dynamic posturography in the diagnosis and treatment of vestibular disorders.

A Health Technology Assessment Report (1998) from the Alberta Heritage Foundation for Medical Research concluded that computerized dynamic posturography is not an established technology in the rehabilitation of vestibular and/or balance deficits associated with stroke, brain injury, and amputation. Dynamic posturography has also been reported to improve the sensitivity of the glycerol test and thus may be useful in the diagnosis and staging of Ménière's disease (Di Girolamo et al, 2001). However, the clinical value of DP for this indication needs to be validated by randomized controlled trials.

Stella et al (2024) noted that surgery for Chiari I malformation (CMI) is indicated when typical clinic-radiological features (syringomyelia, exertional headaches, sleep apnea syndrome, and tetraparesis) are present. Sometimes, patients have atypical complaints suggestive of otolaryngological (ENT) involvement, and it is sometimes difficult for the neurosurgeon to determine if these complaints are related to the CMI. These investigators described postural control patterns in children with CMI using CDP. To the authors' knowledge, this was the 1st study addressing postural instability in pediatric CMI patients. A total of 28 children aged 6 to 17 years with both radiologically confirmed CMI and clinical ENT complaints were included in this study. Subjects were divided into 2 groups -- operated and non-operated patients, based on neurosurgical indication. Epidemiologic and posturographic (CDP-Equitest) findings were compared between the 2 groups, as well as pre- and post-operatively in Group 2. In Group 2 patients, significant improvement of global SOT was found after intervention. When the 3 sensorial aspects of postural control calculated by the system were independently assessed, the greatest improvement was in the vestibular ratio. These researchers also observed an altered CoG pattern ("lateral deviation") in the Group 2 patients, which significantly differed from those in Group 1. Lateral deviation was significantly reduced post-operatively in the Group 2 patients. A correspondence between pre-operative MRI and the side of lateralization on posturography was found in 4 children; however, this could not be regarded as significant due to the low number of patients. The authors concluded that postural control appeared to improve following surgery for CMI in children, mostly due to the improvement in vestibular function. There was also a correspondence between the side of lateral deviation and the side of greatest tonsillar descent on MRI and peri-operatively. Moreover, these researchers stated that further

studies with larger sample sizes are needed to validate these findings and to elucidate the role of CDP in the work-up of CMI patients.

Kohle et al (2024) stated that peripheral neuropathies affects the sensorimotor system, leading to difficulties in walking-related motor tasks and, eventually, falls, which result in functional dependency and reliance on healthcare, especially in the elderly. In a retrospective, case-control study, these investigators examined if peripheral neuropathy is a genuine risk factor for falls in the elderly and if quantification of postural control via CDP is helpful in identifying subjects at risk of falls. A total of 17 elderly with a clinical polyneuropathic syndrome of the lower limbs and converging electrophysiology were compared with 14 older persons without polyneuropathy. All subjects were characterized via quantitative motor and sensory testing, neuropsychological assessment, and self-questionnaires. Video-nystagmography and caloric test excluded vestibulocochlear dysfunction. For further analysis, all subjects were stratified into fallers and non-fallers. Overall, 28 patients underwent CDP for individual fall risk assessment. Regression analyses were carried out to identify risk factors and predictive posturography parameters. Neuropathy is an independent risk factor for falls in the elderly, while no differences were observed for age, gender, weight, frailty, DemTect test, timed "Up & Go" test, and dizziness-related handicap score. In CDP, fallers stepped more often to regain postural control in challenging conditions, while the Rhythmic Weight Shift test showed a lack of anterior-posterior bi-directional voluntary control. The authors concluded that the findings of this study confirmed peripheral neuropathy as a risk factor for falls in the elderly; stepping strategy was often used by fallers to regain postural control. The voluntary control of this coping movement was impaired. Moreover, these researchers stated that further investigations into these parameters' value in predicting the risk of falls in the elderly are needed.

The authors stated that this study had 3 drawbacks. First, these investigators only assessed fall incidence retrospectively; therefore, relied on self-reports. Under-reporting of falls has been reported in geriatric patients. Second, these researchers included multiple neuropathy entities and neurophysiological subtypes in this trial. Especially in immune-mediated neuropathies, fluctuating symptoms had been reported. The authors accounted for this potential confound by including only patients who were clinically stable over the last year and had no treatment change. Third, CDP is a technical assessment that may not be available in most clinical settings, limiting its potential as a screening tool. This trial demonstrated that a sensitive bedside test that evaluates both loss of somatosensory input as well as the voluntary adapting motor control is essential for identification of the elderly at risk for falls.

Cochlear Hydrops Analysis Masking Procedure (CHAMP) Testing for Ménière's Disease

Hong et al (2013) stated that even though it is currently not possible to prove a pathological diagnosis for inner ear disease, acute low-frequency hearing loss (ALFHL) without vertigo could be caused by inner ear hydrops because progression into the clinical spectrum of endolymphatic hydrops (EH) frequently occur among patients with the initial clinical presentation. Therefore, audiological measures representative of inner ear hydrops, such as the cochlear hydrops analysis masking procedure (CHAMP) test, may be used to predict the prognosis of ALFHL without vertigo. To test this hypothesis, these researchers prospectively investigated patients with ALFHL unaccompanied by vertigo and examined whether the CHAMP test generated more useful information for prediction of progression into clinical spectrum of EH compared with other neurotologic parameters. A prospective clinical study of 28 patients who initially presented with ALFHL without vertigo was conducted. Detailed neurotologic findings from pure-tone audiometry, electrocochleography, CHAMP, spontaneous nystagmus, head-shaking nystagmus, vibration-induced nystagmus, the bi-thermal caloric test, and the rotatory chair test were recorded at the time of initial presentation. A regular audiological and clinical examination was conducted until either the last follow-up at the authors' clinic or on the day on which secondary audio-vestibular symptoms occurred. The rates of progression to Ménière's disease (MD) or clinical presentation compatible with isolated cochlear hydrops during the study period were calculated by the log-rank test and relative risk. A receiver operating characteristics curve was plotted to determine the prognostic value of CHAMP. Of 28 patients, 15 (53 %) showed improvement in hearing on pure-tone audiometry. Seven patients (25 %) showed hearing fluctuation and 9 (32 %) developed a vertigo attack during the observation period. Of these, 3 patients experienced both vertigo and a hearing fluctuation. Abnormal results of electrocochleography and neurotologic tests reflecting vestibular ocular reflex on yaw plane were common at the time of diagnosis of ALFHL in

many patients, but these parameters were not associated with an increased risk of progression of clinical spectrum of EH. In contrast, patients with an abnormal complex amplitude ratio (CAR) on CHAMP had a 2.6-fold increased risk of progression to a clinical spectrum of EH (either hearing fluctuation or MD). The hazard ratio of developing MD for patients with normal CAR as compared with those with an abnormal CAR was 0.137 (95 % confidence interval [CI]: 0.03 to 0.57; $p < 0.001$), which indicates an 84.3 % reduced risk of developing MD in those with normal CAR. A CAR value of 0.975 or less indicated the possibility of developing either a hearing fluctuation or vertiginous episode with a sensitivity of 82 % and a specificity of 73 % by receiver operating characteristics curve analysis. The authors concluded that the results of the study suggested that CHAMP measurement may be useful for determining the prognosis of patients with ALFHL without vertigo. A CAR value of 0.975 or less indicated the possibility of developing fluctuating hearing loss or vertigo in patients with ALFHL unaccompanied by vertigo. These findings need to be validated by well-designed studies.

An UpToDate review on “Meniere disease” (Dinces, 2014) states that “The presumed diagnosis of endolymphatic hydrops is based upon clinical symptoms. There is no specific diagnostic test for Meniere disease and a definitive diagnosis can only be made postmortem. The clinical diagnosis in most patients is based upon the history, neurotologic evaluation, and clinical response to medical management. Patients usually have some variable auditory and/or vestibular symptoms for three to five years before they meet the diagnostic criteria for Meniere disease”. It does not mention cochlear hydrops analysis masking procedure (CHAMP) testing for evaluating patients with MD.

Vestibular Rehabilitation

Vestibular rehabilitation is a therapeutic program that utilizes exercises to help regain the sense of balance. Specific movements of the head and body are developed for individual patients with the intent of reducing or eliminating motion provoked or positional sensitivity.

Vestibular rehabilitation (VR) entails the use of specific exercises designed to modify patients' responses to head movement and vestibular stimulation. Vestibular rehabilitation can not prevent the recurrence of active disease, or relieve symptoms without a vestibular origin, or symptoms that are unaffected by position or movement. Patients may be asked to alter head position as well as gaze direction repeatedly, stand for a specified period of time, and perform a specific number of steps with eyes open and shut. Other rehabilitative exercises emphasize balance retraining. Additionally, patients are asked to identify specific positional changes that cause vertigo; the therapy is then designed to have patients execute that position with varying repetitions. After the initial sessions of instruction, patients can usually carry out vestibular rehabilitation exercises at home. For individuals who are uncomfortable to perform the exercises at home, they can do them in an appropriate facility as outpatients.

Vestibular rehabilitation has been used in the treatment of patients with chronic vertigo as a consequence of vestibular dysfunction. It has been reported that patients with chronic peripheral vestibular disorders improved balance and reduced vertigo after 6 weeks of vestibular rehabilitation. Vestibular rehabilitation has also been demonstrated to be beneficial for patients who have undergone ablative vestibular surgery. Vestibular rehabilitation should be performed by a licensed occupational or physical therapist.

The literature indicates that the following groups of patients are generally not good candidates for vestibular rehabilitation:

- Patients with an unstable lesion, usually indicative of a progressive degenerative process (e.g., autoimmune inner ear disease);
- Patients with endolymphatic hydrops, Meniere's disease, or perilymphatic fistula;
- Patients with vertiginous symptoms from a demyelinating disease, epilepsy, or migraine.

In a review on VR for unilateral peripheral vestibular dysfunction, Burton et al (2008) concluded that there is moderate to strong evidence from high-quality randomized trials supporting the safety and effectiveness of this intervention. There is moderate evidence that VR provides a resolution of symptoms in the medium-term. However, there is evidence that for the specific diagnostic group of BPPV, physical (repositioning) maneuvers are more effective in the short-term than exercise-based VR. There is insufficient evidence to discriminate between differing forms of VR.

Vestibular Rehabilitation for Persistent Postural-Perceptual Dizziness

Persistent postural-perceptual dizziness (or PPPD) is a vestibular condition previously referred to as chronic subjective dizziness. Individuals with PPPD don't feel a "room-spinning" type of dizziness or have trouble focusing during head movement like many individuals with vestibular problems experience.

In a systematic review, Trinidad and Goebel (2018) examined the current data on persistent postural-perceptual dizziness (PPPD); and discussed diagnostic criteria and management strategies for this condition with the otologist in mind. Data sources included CINAHL, Embase, PubMed, Medline, PsycINFO, PubMed, Google Scholar. The phrase "persistent postural-perceptual dizziness" and its acronym "PPPD" were used. From 318 articles, 15 were selected for full analysis with respect to PPPD. Most were case-control studies, with 1 consensus paper from the Barany Society available. Overall, the pathophysiology of PPPD remains relatively poorly understood, but is likely to be a maladaptive state to a variety of insults, including vestibular dysfunction and not a structural or psychiatric one. Cognitive behavioral therapy, VR, selective serotonin uptake inhibitors (SSRIs), and serotonin-norepinephrine reuptake inhibitors (SNRIs) all appeared to have a role in its management. The authors concluded that PPPD is useful as a diagnosis for those treating dizziness as it aids in defining a conglomeration of symptoms that could appear otherwise vague and allows for more structured management plans in those suffering from it.

In a review on "Persistent postural-perceptual dizziness", Eldoen and associates (2019) stated that there is substantial evidence that VR is effective in cases of vestibular injury or dysfunction; however, responses may vary depending on the etiology. These investigators noted that VR is an exercise-based treatment consisting of eye, head and body movements designed to stimulate and optimize vestibular compensation. The objective is to reduce the experience of dizziness and imbalance by re-establishing effective and automatic eye-head coordination, reducing anxiety and self-monitoring, increasing fitness, boosting confidence and learning to live with dizziness. The treatment is suitable for anyone who can carry out a daily low-intensity exercise program for 6 to 12 weeks. Many patients can do this independently; others require follow-up by a physiotherapist.

Dunlap and colleagues (2019) examined recent findings of both peripheral and central vestibular disorders and provided insight into evidence related to new rehabilitative interventions. These researchers stated that clinical practice guidelines have recently been developed for peripheral vestibular hypofunction and updated for BPPV. Diagnoses such as PPPD and vestibular migraine are now defined, and there is growing literature supporting the effectiveness of VR as a therapeutic option. As technology advances, virtual reality and other technologies are being used more frequently to augment VR.

Clinicians now have a better understanding of rehabilitation expectations and whom to refer based on evidence in order to improve functional outcomes for individuals living with peripheral and central vestibular disorders.

Mempou and co-workers (2021) noted that visual-vestibular mismatch patients experience PPPD. Previous studies have shown the benefit of VR for visual de-sensitization using gaze stabilization exercises and optokinetic stimulation. In a retrospective study, these researchers examined the benefit of customized VR with visual de-sensitization and virtual reality-based therapy rehabilitation in the management of patients with PPPD. This trial included 100 patients with Situational Characteristic Questionnaire scores of more than 0.9. All patients received virtual reality-based therapy along with usual VR using gaze stabilization exercises with a plain background followed by graded visual stimulation and optokinetic digital video disc stimulation. Patients' symptoms were evaluated before and after VR using the Situational Characteristic Questionnaire, Generalized Anxiety Disorder Assessment-7, Nijmegen Questionnaire and Dizziness Handicap Inventory. There were statistically significant improvements in Situational Characteristic Questionnaire scores, Nijmegen Questionnaire scores and Dizziness Handicap Inventory total score; however, there was a statistically insignificant difference in Generalized Anxiety Disorder Assessment-7 scores. There was a significant positive correlation between post-rehabilitation Situational Characteristic Questionnaire scores and other questionnaire results. The authors concluded that incorporating virtual reality-based therapy with customized VR exercises resulted in significant improvement in PPPD-related symptoms.

An UpToDate review on "Treatment of vertigo" (Furman and Barton, 2021) states that "Most

patients with vertigo prefer to lie with their head still. Vestibular rehabilitation forces them to perform challenging balance exercises with several potential benefits ... Inactivity has secondary negative effects -- Patients may become physically deconditioned, which exacerbates the inadequacy of their postural reflexes. They may also become psychologically deconditioned, sometimes to the point where a "persistent postural-perceptual dizziness" (previously called phobic postural vertigo or chronic subjective dizziness) becomes the greatest obstacle to their recovery. Fear of falling is particularly problematic in older adults after a vestibular event, and it can limit mobility indefinitely without a rehabilitation program".

Furthermore, the Vestibular Disorders Association (VeDA, 2021) provides the following information regarding vestibular balance rehabilitation therapy (VBRT):

Vestibular/balance rehabilitation therapy works to desensitize or habituate patients to motion stimuli. In 2014, the first small study on the efficacy of VBRT specifically for PPPD patients was completed. Its results support previous clinical experience and suggested the following:

- VBRT reduces the severity of vestibular symptoms by 60 % to 80 %, resulting in increased mobility and enhanced daily functioning; and
- VBRT may be effective in reducing anxiety and depression in patients with PPPD; and
- Patients should continue VBRT for 3 to 6 months to receive maximum benefit from the treatments.

Transtympanic Micropressure

Transtympanic micropressure involves implanting a tympanostomy tube between the external ear canal and the middle ear. At the time of treatment, an ear cuff is inserted into the external ear canal and a handheld air-pressure generator (connected to a tabletop air-pressure therapy unit) automatically delivers low-frequency, low-amplitude pressure pulses to the middle ear through the tympanostomy tube. One example of such a device is the Meniett low pressure pulse generator device, portable device developed for the treatment of Ménière's Disease. The Meniett uses positive pressure pulses to the middle ear via the earpiece and ventilation tube to purportedly reduce excess fluid and pressure in the inner ear.

The Meniett device (Medtronic Xomed, Jacksonville, FL) is a local pulsated pressure treatment used for the management of patients with Ménière's disease. It is a portable pressure-pulse generator designed to restore the balance in the hydrodynamics of the inner ear. After a standard ventilation tube is inserted into the tympanum, pressure pulses generated by the Meniett technology are transmitted into the middle ear. The clinical effect occurs as the pulses reach the inner ear. The typical treatment cycle is completed in 5-min sessions, performed 3 times a day. After prescription and training by a physician, patients can treat themselves with the device at home. There is some preliminary evidence that the Meniett device may be effective in treating Ménière's disease.

Odkvist et al (2000) reported that 2-week Meniett treatment resulted in significant improvement concerning frequency and intensity of vertigo, dizziness, aural pressure and tinnitus as indicated on the visual analogue scales questionnaire. Although the findings of this study appear to be promising, its sample size was small (31 in the treatment group and 25 in the control group). Furthermore, there are no long-term follow-up data regarding the effectiveness of this new technology.

Barbara et al (2001) compared the use of ventilation tube (VT) in the middle ear with the combined use of VT and the Meniett device. After a 40-day treatment period, the use of VT alone had a positive effect in 90 % of patients, with either absence (n = 10; 50 %) or marked reduction (n = 8; 40 %) in episodes of vertigo. When Meniett was also applied, stabilization of the positive effect on vertigo was registered, with a concomitant improvement in hearing threshold in 2 patients (10 %). The authors concluded that a longer and more reliable long-term follow-up of this therapeutic approach (VT plus Meniett) is needed.

Gates and Green (2002) also suggested that the Meniett device may be an effective and safe option for people with intractable vertigo from Ménière's disease (n = 10). The findings of these short-term, preliminary descriptive reports of treatment with the Meniett device need to be validated by prospective randomized controlled studies with

larger sample size and adequate follow-up. Furthermore, recent reviews on Ménière's disease (Thai-Van et al, 2001; da Costa et al, 2002) did not mention local pressure treatment as one of the options in treating this condition.

In a small randomized study (n = 62), Gates et al (2004) reported that the Meniett device is safe and effective therapy for treating refractory vertigo in patients with unilateral Ménière's disease. However, this was a short-term clinical study. The investigators agreed that a longer term clinical study was warranted, in part because the difference between treatment and control groups diminished over time. "The significant treatment effect in the treated participants relative to controls over the 4-month trial period diminished over time principally because of apparent spontaneous improvement in the control group. Further assessment over longer periods is needed to better understand the long-term effects of transtympanic micropressure treatment in Ménière's disease." The study by Gates et al has been criticized for failing to use standardized vertigo assessment, for not providing sufficient information on the severity of vertigo in the study population, and for not providing sufficient objective testing data (Reddy and Newlands, UTMB, 2005).

In a small randomized, multi-center, double-blind, placebo-controlled study (n = 40), Thomsen et al (2005) reported that local over-pressure treatment by means of the Meniett device improved statistically significantly the functionality level in patients with Ménière's disease. There was a trend towards a reduction of the frequency of vertiginous attacks that was not significant. However, there were no significant differences between the active and placebo groups in perception of tinnitus, aural pressure, and hearing, before and after the treatment period.

A study by Rajan et al (2005) of the long-term effects of the Meniett device is described as a cross-sectional case study. Well-designed controlled studies are necessary because of the unpredictable natural course of the disease and because of the susceptibility of symptoms to placebo effects.

In a small study (n = 12), Boudewyns et al (2005) reported the effects of the Meniett device in patients with drug-resistant Ménière's disease. With a mean follow-up of 39 months; there was some initial decrease in the frequency of vertigo episodes, but no improvement in functional level, self-perceived dizziness handicap, hearing status or tinnitus. After 1 year, only 2 patients preferred to continue with the therapy. The authors concluded that the Meniett device is unlikely to be helpful in the long-term treatment of severe, drug-resistant Ménière's disease. In addition, the authors pointed out the contrasting findings and recommendations in earlier studies in regard to the patient population with Ménière's disease (e.g., age, stage of disease and severity of vertigo) who are likely to benefit from the treatment.

Gates et al (2006) reported the long-term effectiveness of the Meniett device in patients (n = 58) with classic, unilateral, Ménière's disease unresponsive to traditional medical treatment. The authors concluded that the use of the Meniett device was associated with a significant reduction in vertigo frequency in approximately 2/3 of the subjects, and this improvement was maintained for 2 years. They noted that treatment with the Meniett device is a safe and effective option for people with substantial vertigo uncontrolled by medical therapy. This study was based on an unblinded protocol. Thus, its findings may reflect the effects of treatment, placebo, and/or the natural course of the disease. It should be noted that no objective measurement of hearing was obtained, and most patients indicated that their hearing did not improve with either short-term or long-term use of the Meniett device.

More recently published evidence for the Meniett device consists of small, retrospective case series (Mattox and Reichert, 2008; Dornhoffer and King, 2008; Huang et al, 2009), which are low quality evidence.

Although the Equilibrium Committee of the American Academy of Otolaryngology - Head and Neck Surgery (AAO-HNS, 2008) recommended the use of micropressure therapy (e.g., the Meniett device) as a second level therapy in certain cases of Ménière's disease when medical treatment has failed, the specific criteria for treatment were not listed. Furthermore, this AAO-HNS position does not appear to be supported by a traditional technology assessment of the device/therapy.

In summary, available evidence contain few patients enrolled in randomized, placebo controlled studies, which are critical in differentiating treatment effect to spontaneous improvement that may reflect the natural course of the disorder, including its remissions and recurrences. Furthermore, there are conflicting data regarding which Ménière's

disease patient subsets may benefit from the therapy. Well-designed studies (i.e., larger sample size, randomized, placebo-controlled trials with long follow-up) are needed to establish the safety and effectiveness of the Meniett device for Ménière's disease.

Syed et al (2015) evaluated the effectiveness of the Meniett device in reducing the frequency and severity of vertigo in Ménière's syndrome/disease. The Cochrane Ear, Nose and Throat Disorders Group Trials Register; the Cochrane Central Register of Controlled Trials (CENTRAL); PubMed; EMBASE; CINAHL; Web of Science; BIOSIS Previews; Cambridge Scientific abstracts; ICTRP and additional sources for published and unpublished trials were searched. The date of the last search was May 13, 2014. A total of 4 randomized controlled trials (RCTs) were identified that compared the effectiveness of the Meniett device versus a placebo device in patients with Ménière's 'disease' as defined by the AAO criterion. Two review authors independently assessed study eligibility and risk of bias, and extracted data. The outcome data were dichotomous for all the included trials. The 4 RCTs compared 123 patients with the Meniett device against 114 patients with the placebo device from 4 RCT's over a follow-up period of 2 weeks to 4 months. There was a significant overall 61 % reduction in the frequency of vertigo in both groups [mean no vertigo days per month of 8 to 3]. However, this reduction was not significantly different between the 2 groups in any study or on meta-analysis [mean difference in vertigo free days between Meniett and placebo device of 0.77 days over a 1-month period (95 % confidence intervals [CI]: -0.82 to 1.83) $p = 0.45$]. There was also no substantive data to support a greater reduction in the severity of the vertigo or any other outcome with the Meniett device compared with the placebo device. The authors concluded that no evidence was found to justify the use of the Meniett device in Ménière's syndrome/disease.

In a Cochrane review, van Sonsbeek and colleagues (2015) evaluated the effects of positive pressure therapy (e.g., the Meniett device) on the symptoms of Ménière's disease or syndrome. These investigators searched the Cochrane Ear, Nose and Throat Disorders Group Trials Register; the Cochrane Central Register of Controlled Trials (CENTRAL); PubMed; EMBASE; CINAHL; Web of Science; Cambridge Scientific Abstracts; ICTRP and additional sources for published and unpublished trials. The date of the search was June 6, 2014; RCTs comparing positive pressure therapy (using the Meniett or a similar device) with placebo in patients with Ménière's disease were selected for analysis. The primary outcome was control of vertigo; secondary outcomes were loss or gain of hearing, severity of tinnitus, perception of aural fullness, functional level, complications or adverse effects, and sick days. Two authors independently selected studies, assessed risk of bias and extracted data. They contacted authors for additional data. Where possible, these researchers pooled study results using a fixed-effect, mean difference (MD) meta-analysis and tested for statistical heterogeneity using both the χ^2 test and I^2 statistic. This was only possible for the secondary outcomes loss or gain of hearing and sick days. They presented results using forest plots with 95 % CI. These investigators included 5 randomized clinical trials with 265 participants. All trials were prospective, double-blind, placebo-controlled RCTs on the effects of positive pressure therapy on vertigo complaints in Ménière's disease. Overall, the risk of bias varied: 3 out of 5 studies were at low risk, 1 was at unclear risk and 1 was at high risk of bias. For the primary outcome, control of vertigo, it was not possible to pool data due to heterogeneity in the measurement of the outcome measures. In most studies, no significant difference was found between the positive pressure therapy group and the placebo group in vertigo scores or vertigo days. Only 1 study, at low risk of bias, showed a significant difference in 1 measure of vertigo control in favor of positive pressure therapy. In this study, the mean visual analog scale (VAS) score for vertigo after 8 weeks of treatment was 25.5 in the positive pressure therapy group and 46.6 in the placebo group (MD -21.10, 95 % CI: -35.47 to -6.73; scale not stated - presumed to be 0 to 100). For the secondary outcomes, these investigators carried out 2 pooled analyses. They found statistically significant results for loss or gain of hearing.

Hearing was 7.38 decibels better in the placebo group compared to the positive pressure therapy group (MD) (95 % CI: 2.51 to 12.25; 2 studies, 123 participants). The severity of tinnitus and perception of aural fullness were either not measured or inadequate data were provided in the included studies. For the secondary outcome functional level, it was not possible to perform a pooled analysis. One included study showed less functional impairment in the positive pressure group than the placebo group (AAO-HNS criteria, 1- to 6-point scale: MD -1.10, 95 % CI: -1.81 to -0.39, 40 participants); another study did not show any significant results. In addition to the pre-defined secondary outcome measures, these researchers included sick days as an additional outcome measure, as 2 studies used this outcome measure and it is a complementary measurement of impairment due to Ménière's disease. They did not find a statistically significant difference in sick days. No

complications or adverse effects were noted by any study. The authors concluded that there is no evidence, from 5 included studies, to show that positive pressure therapy is effective for the symptoms of Ménière's disease. There is some moderate quality evidence, from 2 studies, that hearing levels are worse in patients who use this therapy. The positive pressure therapy device itself is minimally invasive. However, in order to use it, a tympanostomy tube (grommet) needs to be inserted, with the associated risks. These include the risks of anesthesia, the general risks of any surgery and the specific risks of otorrhea and tympano-sclerosis associated with the insertion of a tympanostomy tube.

Brainstem Auditory Evoked Potentials for Evaluation of Vertigo

Ji and Zhang (2014) examined the relationship and changes of cervical magnetic resonance imaging (MRI), transcranial Doppler (TCD), and brainstem auditory evoked potential (BAEP) in patients with "isolated" vertigo. The relationship and changes of cervical MRI, TCD and BAEP were investigated in 125 patients with "isolated" vertigo and 100 healthy controls. There were statistically significant differences between 2 groups for overall abnormalities of TCD ($X^2 = 61.96$, $p < 0.01$), BAEP ($X^2 = 97.99$, $p < 0.01$), and cervical MRI severity scale ($Z = -8.71$, $p < 0.01$). In vertigo group, results showed significant correlations between TCD and cervical MRI, TCD and BAEP as well. And analysis on TCD pulsatility index and some items of BAEP demonstrated positive linear correlations. There were no statistical differences or correlations in control group. The authors concluded that TCD was a sensitive method of "isolated" vertigo screening. They stated that a combined test protocol of cervical MRI, TCD and BAEP has superiorities to assess "isolated" vertigo.

The authors stated that this study had several drawbacks. First, the number of patients in this study was relatively small. Second, these researchers did not evaluate some vascular risk factors, including tobacco smoking, diabetes, hypertension, and so on. Third, it was not a randomized control study, which might have resulted in clinical bias. These investigators pointed out that, image changes, pathological results, and a long term follow-up is needed to support the conclusions of this study, and to overcome inherent defects of cross-sectional study.

He and colleagues (2015) used high (49/s) and low (9/s) stimulation rates of the BAEP to examine the possible mechanism responsible for BPPV. A total of 81 patients (55 women and 26 men, mean age $\pm$ SD = 54.6 ± 15.0 years) with idiopathic BPPV, as well as 106 control subjects (70 women and 36 men, mean age $\pm$ SD = 51.2 ± 16.3 years) participated in the study. The results of high (49/s) and low (9/s) stimulation rates of the BAEP test were compared and analyzed. The difference in BAEP wave I peak latencies between low and high stimulation rate (DPL I) and BAEP wave I peak latency in high stimulation (HPL I) of affected ears (0.24 ± 0.14 and 1.91 ± 0.21 ms) in BPPV patients were significantly prolonged when compared with the controls (0.10 ± 0.08 and 1.76 ± 0.18 ms) and unaffected ears (0.12 ± 0.10 and 1.82 ± 0.21 ms) ($p < 0.001$). The abnormal rate of DPL I in the affected ear (52/83, 62.65 %) was significantly higher than that in the unaffected ear (7/79, 8.86 %) and the normal left ear (4/106, 3.77 %). The authors concluded that these findings suggested that ischemia of the inner ear might be one of the causes of BPPV and that DPL I may be used to assess the ischemic degree in subjects over 20 years of age.

An UpToDate review on "Evaluation of the patient with vertigo" (Furman and Barton, 2018) states that "Brainstem auditory evoked potentials (BAEPs) have a 90 to 95 % sensitivity for detecting acoustic neuromas. Any type of sensorineural hearing loss will disrupt the pattern of sound passing from the cochlea to the brainstem; abnormal results are therefore non-specific. BAEPs are no longer routinely used in the evaluation of patients with vertigo or suspected vestibular schwannoma".

Biodex BioSway Balance System for Balance Assessment

Parsa et al (2019) noted that balance disorders are considered to be a serious clinical manifestation after stroke; thus, use of a quantitative method appears essential for evaluation of stroke patients' balance performance. A fundamental step would be the approval of the efficiency of the measurement instruments. These investigators examined correlations between balance assessment as examined by Biodex Stability System (BSS) and the clinical Berg Balance Scale (BBS) in post-stroke hemiparesis. A total of 25 stroke survivors and 25 healthy age-sex matched subjects were recruited. Participants were evaluated using BSS during 3 days, with a 24-hour interval. The high inter-class correlation coefficient (ICC) values showed that the system was reliable enough to

continue the study. The clinical evaluation was performed by the standard BBS. There was a significant moderate negative correlation between the Biodex overall indices and BBS scores in the stroke groups (ravg = -0.68) and in the healthy cohort (ravg = -0.55). Furthermore, a significant moderate negative correlation was observed between the Biodex antero-posterior stability indices and BBS scores in the stroke groups (ravg = -0.67) and in healthy cohort (ravg = -0.55). The correlation between the Biodex mediolateral stability indices and BBS scores was moderate-to-low in the stroke and healthy groups (ravg = -0.67 and -0.39, respectively). The authors concluded that moderate negative correlation between the stability indices of the Biodex Stability System and BBS scores indicated that dynamic balance status of the participants partially reflected their functional balance status.

Dewan et al (2019) stated that the Biodex Biosway Balance System and SWAY Balance Mobile smartphone application (SBMA) are portable instruments that evaluate balance function with force plate and accelerometer technology, respectively. The validity of these indirect clinical measures of postural sway merits investigation. In a cross-sectional; repeated measures study, these investigators examined the concurrent validity of standing postural sway measurements by using the portable Biosway and SBMA systems with kinematic measurements of the whole-body center of mass (COM) derived from a gold-standard reference, a motion capture system. A total of 40 healthy young adults (21 females, 19 males) participated in this study. Subjects carried out 10 standing balance tasks that included combinations of standing on 1 or 2 legs, with eyes open or closed, on a firm surface or foam surface and voluntary rhythmic sway. Postural sway was measured simultaneously from SBMA, Biosway, and the motion capture system. The linear relationships between the measurements were analyzed. Significant correlations were found between Biosway and COM velocity for both progressively challenging single- and double-leg stances ($\tau_b = 0.3$ to 0.5 , $p < 0.01$ to < 0.0001). SBMA scores and COM velocity were significantly correlated only for single-leg stances ($\tau_b = -0.5$ to -0.6 , $p < 0.0001$). SBMA scores had near-maximal values with zero to near-zero variance in double-leg stances, indicating a ceiling effect. The authors concluded that the force plate-based Biodex Biosway was valid for evaluating standing postural sway for a wide range of test conditions and challenges to standing balance, whereas an accelerometer-based SWAY Balance smartphone application was valid for evaluating postural sway in progressively challenging single-leg stance but was not sensitive enough to detect lower-magnitude postural sway changes in progressively challenging double-leg stances. Moreover, these researchers stated that insights from this study, and further investigations in clinical populations, will be useful in the selection of a portable objective clinical balance assessment instruments.

The authors stated that a drawback of this study was using a belt to hold the smartphone with SBMA that was tied around the chest at the mid-sternum level, which was different from the SBMA default procedure of holding the device with both hands at the mid-sternum level. This procedural alteration was needed to minimize errors in measurement from the movements of device with the subject's hand and to standardize the positioning of the device in all subjects to prevent any inconsistencies due to technique. Another drawback was the generalizability of this study, which included the use of asymptomatic subjects aged 20 to 34 years; therefore, extrapolation to healthy subjects in different age groups or to patient populations with balance impairments may not be appropriate. Areas for further investigations include the assessment of validity of the 2 instruments in these populations, which could provide more generalizable information regarding the validity of portable balance assessment instruments that use accelerometers versus force plates.

Miner et al (2020) noted that current tools for sideline assessment of balance following a concussion may not be sufficiently sensitive to identify impairments, which may place athletes at risk for future injury. Quantitative field-expedient balance assessments are becoming increasingly accessible in sports medicine and may improve sensitivity to enable clinicians to more readily detect these subtle deficits. In a cross-sectional, cohort study, these researchers determined the validity of the postural sway assessment on the Biodex BioSway compared with the gold standard NeuroCom Smart Equitest System. A total of 49 healthy adults (29 females: 24.34 [2.45] years of age, height of 163.65 [7.57] cm, body weight of 63.64 [7.94] kg; 20 males: 26.00 [3.70] years of age, height of 180.11 [7.16] cm, body weight of 82.97 [12.78] kg). Subjects completed the modified clinical test of sensory interaction in balance on the Biodex BioSway with 2 additional conditions (head shake and firm surface; head shake and foam surface) and the Sensory Organization Test (SOT) and Head Shake Sensory Organization Test (HSSOT) on the NeuroCom Smart Equitest. Main outcome measures were inter-class correlation coefficient (ICC) and Bland-Altman limits of agreement for Sway Index, equilibrium ratio, and area of 95 %

confidence ellipse. Fair-good reliability (ICC = 0.48 to 0.65) was demonstrated for the stance conditions with eyes open on a firm surface. The Head Shake Sensory Interaction and Balance Test condition on a firm surface resulted in fair reliability (ICC = 0.50 to 0.59). These researchers observed large ranges for limits of agreement across outcome measures, indicating that the systems should not be used interchangeably. The authors observed fair reliability between BioSway and NeuroCom, with better agreement between systems with the assessment of postural sway on firm/static surfaces. However, the agreement of these systems may improve by incorporating methods that mitigate the floor effect in an athletic population (e.g., including a head shake condition). These investigators stated that the Biodex BioSway may provide a surrogate field-expedient measurement tool.

In a cross-sectional study, Karartı et al (2021) determined cut-off scores for the Biodex Balance System (BBS) and examined if they could be used to discriminate older people with non-specific low-back pain (NSLBP) with poor postural performance from those with good postural performance. This trial included 52 subjects with NSLBP older than 65 years. One level of stability (level 5; ICC greater than or equal to 0.70) and 2 conditions (eyes open and eyes closed) were selected for the testing procedure. Anterior-posterior stability index (APSI), medial-lateral stability index (MLSI), and overall stability index (OSI) scores were calculated. Subjects were classified into 2 groups: high-risk of falling and low-risk of falling. Both the receiver operating characteristic and the area under the curve were used to determine the best BBS cut-off values. Binary logistic regression analysis was used to examine the ability of BBS scores to predict risk of falling. BBS cut-off scores in the eyes-open condition (APSI = 2.60, MLSI = 1.95, OSI = 2.95) and eyes-closed condition (APSI = 3.05, MLSI = 2.17, OSI = 3.25) were found to be sensitive and specific in determining postural performance. Subjects with index values lower than the cut-off scores had, respectively, 6.42, 4.20, and 3.72 times lower risk of falling in the eyes-open condition and 3.33, 5.50, and 3.00 times lower risk of falling in the eyes-closed condition. The predictive characteristics of the models for risk analysis were excellent and good-to-excellent. The authors concluded that the findings of this study showed that BBS cut-off scores were sensitive and specific in distinguishing between poor and good postural performance in older people with NSLBP.

Posterior Semicircular Canal Occlusion for the Treatment of Benign Paroxysmal Positional Vertigo

Zhu and colleagues (2015) noted that several studies have suggested that semicircular canal occlusion is safe and effective for treating intractable posterior semicircular BPPV (PSC-BPPV), and adverse effects of canal occlusions for intractable horizontal semicircular BPPV (HSC-BPPV) were rarely reported. In a retrospective study, these researchers examined the efficacy of semicircular canal occlusion for intractable HSC-BPPV with at least 2 years of follow-up. From 2000 to 2011, a total of 3 women (average age of 60 ± 6.9 years), with a diagnosis of HSC-BPPV refractory to head-shake and barbecue roll maneuver, underwent semicircular canal occlusion treatment in the authors' hospital. The supine roll test was performed to diagnose HSC-BPPV and examine the treatment efficacy. All patients with intractable HSC-BPPV had completed resolution of their positional vertigo following semicircular canal occlusion with a negative supine roll test. All patients reported transient post-operative disequilibrium, nausea, and vomiting, which resolved within 2 weeks. Furthermore, 1 patient (33.3 %) had transient tinnitus, which resolved after 4 months. There were no other significant long-term complications. The authors concluded that semicircular canal occlusion appeared to be a safe and well-tolerated treatment modality for intractable HSC-BPPV; however, further studies with large sample sizes are needed to confirm these preliminary findings.

Maas and associates (2020) studied the effect of posterior canal occlusion for intractable posterior canal BPPV on vertigo and examined the risk of loss of auditory or vestibular function. These researchers carried out a systematic literature search according to the PRISMA statement on PubMed, the Cochrane Library, Embase, Web of Science, and CINAHL. The last search was conducted in June 2018. Cohort studies with original data and case reports describing more than 5 cases were included if they analyzed the effect of posterior semicircular canal obliteration in adults with intractable posterior BPPV on vertigo. Two authors screened titles and abstracts for eligibility. The 1st author screened full texts and analyzed the data. A total of 8 retrospective studies met the eligibility criteria. The quality of all individual studies was rated fair. Canal occlusion was carried out on 196 patients. All studies reported complete resolution of BPPV in all patients (100 %). Among post-operatively

tested patients, total loss of auditory function and vestibular function was reported in 2 of 190 (1 %) and 9 of 68 (13 %), respectively. The authors concluded that posterior semicircular canal plugging resulted in 100 % resolution of BPPV in patients with intractable BPPV in all studies; however, the strength of evidence was weak. These researchers stated that potential serious complications, such as deafness and loss of vestibular function, should be taken into account.

Furthermore, an UpToDate review on “Benign paroxysmal positional vertigo” (Barton, 2021) states that “BPPV is intractable in a very small number of patients. Surgical treatments may be considered in patients who are disabled by their symptoms. Surgical occlusion of the posterior canal with bony plugs is one option; success rates of approximately 90 % have been reported for this procedure in uncontrolled reports. This surgery renders the posterior canal permanently nonfunctional; transient postoperative hearing loss and dizziness are very common. Persistent hearing loss occurs in less than 5 %; hence, impaired hearing in the other ear is a contraindication to this procedure”. However, “surgical occlusion” is not listed in the “Summary and Recommendations” section of this review.

Cervical and Ocular Vestibular Evoked Myogenic Potentials

The American Academy of Neurology (Fife, et al., 2017) systematically reviewed the evidence and made recommendations with regard to diagnostic utility of cervical and ocular vestibular evoked myogenic potentials (cVEMP and oVEMP, respectively). Four questions were asked: Does cVEMP accurately identify superior canal dehiscence syndrome (SCDS)? Does oVEMP accurately identify SCDS? For suspected vestibular symptoms, does cVEMP/oVEMP accurately identify vestibular dysfunction related to the saccule/utricle? For vestibular symptoms, does cVEMP/oVEMP accurately and substantively aid diagnosis of any specific vestibular disorder besides SCDS? The guideline panel identified and classified relevant published studies (January 1980-December 2016) according to the 2004 American Academy of Neurology process. The AAN made a Level C positive recommendations that clinicians may use cVEMP stimulus threshold values to distinguish SCDS from controls (2 Class III studies) (sensitivity 86%-91%, specificity 90%-96%). Corrected cVEMP amplitude may be used to distinguish SCDS from controls (2 Class III studies) (sensitivity 100%, specificity 93%). Clinicians may use oVEMP amplitude to distinguish SCDS from normal controls (3 Class III studies) (sensitivity 77%-100%, specificity 98%-100%). oVEMP threshold may be used to aid in distinguishing SCDS from controls (3 Class III studies) (sensitivity 70%-100%, specificity 77%-100%). The AAN guideline also included the following Level U statements: Evidence is insufficient to determine whether cVEMP and oVEMP can accurately identify vestibular function specifically related to the saccule/utricle, or whether cVEMP or oVEMP is useful in diagnosing vestibular neuritis or Ménière disease. The guideline also included the following Level C negative recommendations: It has not been demonstrated that cVEMP substantively aids in diagnosing benign paroxysmal positional vertigo, or that cVEMP or oVEMP aids in diagnosing/managing vestibular migraine.

Lee et al (2017) noted that vestibular-evoked myogenic potentials (VEMPs) can be abnormal in patients with idiopathic recurrent spontaneous vertigo. These researchers examined if abnormal cervical VEMPs (cVEMPs) can predict evolution of isolated recurrent vertigo into Meniere's disease (MD). They had followed-up 146 patients with isolated recurrent vertigo and an evaluation of cVEMPs for 0 to 142 months [median of 6, inter-quartile range (IQR) = 0 to 29] at the Dizziness Clinic of Seoul National University Bundang Hospital from June 2003 to May 2014. These investigators defined the variables associated with a progression into MD and calculated cumulative progression rates. Among the 94 patients with recurrent vertigo and abnormal cVEMPs, 18 (18/94, 19 %) showed an evolution into MD while only 2 of the 50 (4 %) patients with normal cVEMPs evolved into MD during the follow-up ($p = 0.01$). The interval between onset of vertigo and development of cochlear symptoms ranged from 1 month to 13.6 years (median of 3 years, IQR = 0.5 to 4.5 years). Overall, pure tone audiometry (PTA) threshold at 0.25-kHz [hazard ratio (HR) = 1.1, 95 % confidence interval (CI): 1.0 to 1.2] and abnormalities of cVEMPs (HR = 5.6, 95 % CI: 1.3 to 25.5) were found to be significantly associated with a later conversion into MD. The cumulative progression rate was 12 % (95 % CI: 5 to 18) at 1 year, 18 % (8 to 26) at 2 years, and 22 % (11 to 32) at 3 years. The authors concluded that abnormal cVEMPs may be an indicator for evolution of isolated recurrent vertigo into MD. Patients with isolated recurrent vertigo may be better managed conforming to MD when cVEMPs are abnormal.

Semmanaselvan et al (2019) stated that VEMP abnormalities in individuals with benign

paroxysmal positional vertigo (BPPV) are often reported to be associated with utricle and saccule degeneration. These researchers evaluated the frequency of VEMP abnormalities using VEMPs in individuals with posterior canal BPPV after Epley's maneuver. A total of 36 individuals (36 ears) with definite posterior canal BPPV and 36 healthy controls were considered for the present study. All subjects underwent otoscopic examination, Dix-Hallpike maneuver to diagnose posterior canal BPPV. Further audiological evaluation including PTA was performed to rule out vestibular disorders associated with hearing loss. Epley's maneuver was performed on all individuals with BPPV by an experienced otorhinolaryngologist. Cervical and ocular VEMP were used to examine the saccule and utricle functions following Epley's maneuver. Cervical VEMP (cVEMP) and ocular VEMP (oVEMP) abnormalities were observed in 8/36 (22.22 %) and 18/36 (50 %) affected ears with BPPV, respectively. Cervical VEMP responses were reduced in amplitude among 1/36 (2.77 %) and absent in 7/36 (19.44 %) of affected ears with BPPV. Ocular VEMP responses were reduced in amplitude on 11/36 (30.55 %), followed by absent responses in 5/36 (13.88 %) ears with BPPV; 2 patients with posterior canal BPPV i.e., 4/64 (5.55 %) ears had bilateral absence of oVEMP responses. Two ears with BPPV 2/36 (5.55 %) had absence of both cVEMP and oVEMP responses in BPPV affected ear. T-test showed significant difference ($p < 0.01$) in the amplitude of oVEMP among posterior canal BPPV individuals when compared to cVEMP. The authors concluded that the findings of this study highlighted individuals with posterior canal BPPV may have otoconia dislodgement or macular degeneration of utricle, saccule, both utricle and saccule unilaterally, or bilaterally. These researchers stated that VEMP may be useful in evaluating degeneration of both otolith organs associated with BPPV.

Xu et al (2019) examined the diagnostic value of VEMP (cVEMP and oVEMP), caloric test, and cochlear electrogram (EcochG) in patients with Meniere's disease (MD) and non-MD. A total of 64 patients (64 ears) with unilateral MD were enrolled in the study group (MD group), and 127 cases (254 ears) of non-MD patients as non-MD group, including vertigo migraine in 40 cases, BPPV in 48 cases, benign recurrent vertigo in 13 cases, vestibular paroxysmia in 3 cases, vestibular neuritis in 5 cases and other undiagnosed vertigo in 18 cases. Both groups underwent cVEMP, oVEMP, caloric test and EcochG. Medcalc software was used to draw ROC curve of EcochG and calculate the area under curve (AUC), Jordan index and optimal diagnostic cut-off points. The cut-off point was the point of -SP/AP, then the sensitivity, specificity, positive predictive value (PPV), negative predictive value (NPV), and diagnostic accuracy of cVEMP, oVEMP, caloric test and EcochG in MD group and non-MD group were evaluated. The AUC of EcochG ROC curve was 0.74, the Jordan index was 0.47 and the cut-off point was 0.4. The sensitivity and specificity of cVEMP (62 % and 68 %), oVEMP (61 % and 53 %), and caloric test (53 % and 57 %) were all below EcochG (65 % and 78 %). The PPV of EcochG was the highest (61.9 %), the NPV of cVEMP was highest (87.5 %). The diagnostic accuracy of EcochG was highest (74 %), followed with cVEMP (67 %), oVEMP (55 %) and caloric test (56 %). The authors concluded that compared with the vestibular function tests, the sensitivity, specificity, diagnostic accuracy and NPV were all higher in EcochG, and the diagnostic benefit can be maximized when -SP/AP value greater than 0.4. Thus, the value of single vestibular function examination in the diagnosis of Meniere's disease is limited. The diagnosis of MD still requires a comprehensive evaluation in combination with medical history, audiological tests and vestibular function examinations.

Gunes et al (2020) noted that cVEMP measurements still do not have standard normative values in posterior canal BPPV. These researchers compared cVEMP recordings obtained with different stimuli applied in 2 different intensities in posterior canal BPPV patients. A total of 34 patients with unilateral posterior canal BPPV were included in the patient group. In cVEMP recordings obtained with different stimulus intensity [95 dB HL and 105 dB HL] and different stimuli [tone-burst cVEMP (T-cVEMP) and click cVEMP (C-cVEMP)]. When the C-cVEMP and T-cVEMP findings were compared in the patient group, differences were observed only in peak-to-peak p1-n1 amplitude values in the measurements performed with 95 dB stimulus; however, T-cVEMP measurements performed with 105-dB stimulus showed that both p1 and n1 latency values were longer and peak-to-peak p1-n1 amplitude values were higher than C-cVEMP measurements. The authors recommended using pure tone-burst stimulus for measurements with 105-dB HL in cVEMP evaluations they would perform in posterior BPPV patients. Both stimulants can be used when 95-dB HL stimuli is used.

Balanceback Intuitive VNG Device for the Management of Balance Disorders

The BalanceBack VNG device, cleared for marketing via the 510(k) process, is intended for recording, viewing, and analyzing eye movements in support of identifying balance

disorders in human subjects. The VNG is intended for use only by trained physicians or clinicians in an appropriate doctor's office or health care facility. This device provides no diagnoses; nor does it provide diagnostic recommendations.

There is a lack of evidence to support the use of the Balanceback Intuitive VNG device for the management of balance disorders.

Balance Tracking System (BTrackS) for Evaluation of Risk of Falling

The Balance Tracking System (BTrackS) employs standard force plate technology to provide balance testing and training. However, there is insufficient evidence to support the use of BTrackS for evaluation of risk of falling. The majority of the published evidence is from one group of investigators (Goble and colleagues, 2017, 2018, 2020, and 2021)

Goble et al (2017) noted that atypically high postural sway measured by a force plate is a known risk factor for falls in older adults. Furthermore, it has been shown that small, but significant, reductions in postural sway are possible with various balance exercise interventions. In the present study, a new low-cost force-plate technology called the Balance Tracking System (BTrackS) was employed to examine postural sway of the elderly before and after 90 days of a well-established exercise program called Geri-Fit. Results showed an overall reduction in postural sway across all participants from pre- to post-intervention. However, the magnitude of effects was significantly influenced by the amount of postural sway demonstrated by individuals before Geri-Fit training. Specifically, more participants with atypically high postural sway pre-intervention experienced an overall postural sway reduction. These reductions experienced were typically greater than the minimum detectable change statistic for the BTrackS Balance Test. The authors concluded that the findings of this study suggested that BTrackS is an effective means of identifying the elderly with elevated postural sway, who are likely to benefit from Geri-Fit training to mitigate fall risk.

The authors stated that although sample size in this study was relatively small ($n = 25$; 6 men and 19 women), the effects were large enough to show significant differences across testing time-points and between LOWER and HIGHER postural sway groups. This study also lacked a "true" control group of older adults where postural sway assessments were made along a similar time course to those individuals in the present study.

Levy et al (2018) stated that falls are the leading cause of disability, injury, hospital admission, and injury-related death among the elderly. Balance limitations have consistently been identified as predictors of falls and increased fall risk. Field measures of balance are limited by issues of subjectivity, ceiling effects, and low sensitivity to change. The gold standard for measuring balance is the force plate; however, its field use is untenable due to high cost and lack of portability. Thus, there is a critical need for valid objective field measures of balance to accurately evaluate balance and identify limitations over time. These researchers examined the concurrent validity and 3-day test-retest reliability of the Balance Tracking System (BTrackS) in community-dwelling older adults. Minimal detectable change values were also calculated to reflect changes in balance beyond measurement error. Postural sway data were collected from community-dwelling older adults ($n = 49$, mean [SD] age = 71.3 [7.3] years) with a force plate and BTrackS in multi-trial eyes open (EO) and eyes closed (EC) static balance conditions. Force sensors transmitted BTrackS data via a USB to a computer running custom software. Three approaches to concurrent validity were taken including calculation of Pearson product moment correlation coefficients, repeated-measures ANOVAs, and Bland-Altman plots. Three-day test-retest reliability of BTrackS was examined in a 2nd sample of 47 community-dwelling older adults (mean [SD] age = 75.8 [7.7] years) using intra-class correlation coefficients and MDC values at 95 % CI (MDC95) were calculated. BTrackS demonstrated good validity using Pearson product moment correlations ($r > 0.90$). Repeated-measures ANOVA and Bland-Altman plots indicated some BTrackS bias with center of pressure (COP) values higher than FP COP values in the EO (mean [SD] bias = 4.0 [6.8]) and EC (mean [SD] bias = 9.6 [12.3]) conditions. Test-retest reliability using intra-class correlation coefficients (ICC2.1 was excellent (0.83) and calculated MDC95 for EO (9.6 cm) and EC (19.4 cm) and suggested that postural sway changes of these amounts were meaningful. BTrackS showed some bias with values exceeding force plate values in both EO and EC conditions. Excellent test-retest reliability and resulting MDC95 values indicated that BTrackS has the potential to identify meaningful changes in balance that may warrant intervention. The authors concluded that BTrackS is an objective measure of balance that can be used to monitor balance in community-dwelling older adults over time. It can reliably identify changes

that may require further attention (e.g., fall-prevention strategies, declines in physical function) and showed promise for evaluating intervention effectiveness in this growing segment of the population.

Goble and Baweja (2018) noted that the BTrackS is a new force plate that alleviates these barriers and has potential for widespread use. These researchers provided important normative data for the BTrackS Balance Test of postural sway that improves its translational value to the field of gerontology. BTrackS Balance Test postural sway results were accumulated from 6,280 community-dwelling individuals across the adult lifespan. Data were assessed for effects of age, sex and body size. Stratified percentile rankings were then calculated. BTrackS Balance Test results were dependent on age and sex, but not body size. Percentile rankings were, therefore, determined across various age groups for men and women separately, with no consideration of participant body size. A novel interaction was found between the age and sex factors, suggesting enhanced postural sway ability for women that becomes more pronounced with older age. The authors concluded that BTrackS is an emerging clinical tool that, combined with the percentile ranking normative data in the present study, translates into the field as an objective means of determining abnormalities in postural sway. Such abnormalities have previously been associated with a number of poor clinical outcomes including, importantly, increased fall risk. Fortunately, once identified, postural sway deficits can be treated by using exercise-based training interventions. To this extent, it is worth noting that a recent study (Goble et al, 2017) showed that BBT (i.e., postural sway) reductions were possible for older adults using a 90-day resistance training intervention.

The authors stated that this study had several limitations. First, data were collected at multiple sites, with no direct oversight by the authors. This was necessary to obtain such a large sample size, which it was believed is the 2nd largest postural sway dataset ever published. Second, BBT data relied on the self-reported adherence of test sites to implementing the standardized BBT protocol correctly. This might have been more difficult for the elderly tested, and it was unlikely that 100 % compliance was achieved across the entire sample. Furthermore, it is unknown to what extent performance feedback was provided to the individuals during testing or after a given trial. Third, there was a sample imbalance across the various age/sex groups compiled. Although 100 or more individuals were tested in each age/sex group to allow a true percentile ranking to be determined, some groups were "over-represented" with 1,000 or more samples collected. In the future, gathering a more balanced sample of BBT results across age and sex groups might serve to correct small errors in the present values. In addition, geographic demographic factors were also worth considering in future efforts to obtain normative BBT data. The present sample might not have been truly representative of the global population at large, as testing sites were primarily located in larger North American cities that might have had a greater percentage of higher socioeconomic status individuals.

In a mixed-methods, case-series study, Cappleman and Thiamwong (2020) evaluated fear of falling (FOF) in community-dwelling older adults using subjective and objective measure; and examined older adults' perceptions of FOF assessments and interventions. This trial consisted of quantitative data collection by objective measures including the BTrackS Balance Test (BBT) and a dynamometer to assess physiological fall risk, and in-depth interviews from 4 older adults in Orlando (FL). A single FOF Scale and Falls Efficacy Scale-International (FES-I) were used to assess FOF. To combine quantitative and qualitative data, a case-specific analysis was used and followed by a cross-case analysis to gain a more comprehensive understanding of FOF. These researchers found an incongruent FOF with physiological fall risk. A total of 4 themes emerged: First, fluctuating definitions of "fear" contributed to difficulty in assessments and interventions. Second, fundamental assessments for FOF were missing. Third, feedback from an objective measure was valuable. Fourth, family experiences with FOF drove personal interventions. The authors concluded that the integrated viewpoints from quantitative and qualitative data suggested a need for FOF assessment based on older adults' perceptive and physiological measures.

Haworth et al (2020) stated that unconstrained limits of stability (LoS) assessment revealed aspects of volitional postural sway control that are inaccessible by other means. Prior versions of this assessment included instructions to sway towards pre-defined targets; and may not capture the full capability of the individual. These researchers sought to establish the test-retest reliability of a novel Balance Tracking System (BTrackS) limits of stability protocol. Volitional sway area was determined

during unconstrained trials, where participants were instructed to explore their ability to sway towards the perimeter of their base of support. Visual feedback was provided via computer monitor. A total of 40 healthy young adults (mean age of 20.2 ± 1.3 , 15 males, 25 females) participated in this study. Trials were collected in 3 sessions, repeated at the same time of the same day, with 1 week between. Reliability was assessed using ICC, considering the total area of sway as well as quadrant level area. Reliability was moderate between the 1st and 2nd session (0.583), and much higher (0.921) between the 2nd and 3rd session. The quadrant level reliability was poor-to-excellent (0.183 to 0.791), with similar trends between the 3 sessions. The authors concluded that these findings indicated that the novel limits of stability test were reliable; however, it is recommended that a practice trial be carried out before baseline establishment. Moreover, these researchers stated that further investigation is needed to examine the degree of malleability due to interventions (e.g., stretching, strengthening) or mood (e.g., pain, motivation). The BTrackS LoS test, in isolation or combined with static sway area, may afford new insights into context-dependent and goal-specific effectors of postural sway regulation. The main drawback of this study was that it was carried out in healthy young volunteers; thus, the implications of these findings in the setting of evaluation of risk of falling in the elderly need to be further investigated.

Goble et al (2021) noted that postural control is important for body sway control and is subserved by 3 sources of sensory feedback (i.e., vision, proprioception and vestibulation). A method for determining the relative contribution of each sensory feedback source to postural control is the modified clinical test of sensory integration and balance for the balance tracking system (BTrackS). However, this method has not yet been evaluated for test-retest reliability. These researchers examined the test-retest reliability of the modified clinical test of sensory integration and balance protocol for the BTrackS across multiple time intervals. A total of 3 groups of healthy young adults performed the BTrackS modified clinical test of sensory integration and balance protocol 4 times separated by either 1 day, 1 week or 1 month. Within each time duration group, and condition, differences in total center of pressure path length were determined from one test session to the next and intra-class correlation coefficient categorizations were made. In all but 1 case, no significant difference in performance was observed from one testing session to the next. The one significant difference found was a decrease in total center of pressure path length from day 1 to day 2 in the vestibular condition of the group tested daily. Intra-class correlation coefficient results largely indicated fair-good reliability across time durations and test conditions. The authors concluded that the findings of this trial largely supported using the BTrackS mCTSIB as a relatively reliable means of probing sensory contributions to balance performance across multiple time durations. However, care should be taken when interpreting a vestibular condition result if taken over a day-to-day duration, as this may be prone to a practice effect. To this point, inclusion of a practice trial for the vestibular condition might be recommended in a future version of the protocol. This would allow an opportunity for the participant to accommodate to the difficulty associated with eyes closed standing on a compliant foam surface. The main drawback of this study was that it was carried out in healthy young volunteers; thus, the implications of these findings in the setting of evaluation of risk of falling in the elderly need to be further investigated.

Chen et al (2021) stated that quantitative assessment is important for the evaluation of human postural balance. The force plate system is the key quantitative balance assessment method. These researchers examined the concepts in balance assessment and analyzed the experimental conditions, parameter variables, and application scope based on force plate technology. They noted that as there is a wide range of balance assessment tests and a variety of commercial force plate systems to choose from, there is room for further improvement of the test details and evaluation variables of the balance assessment. The recommendations presented in this article were the foundation and key part of the postural balance assessment; these recommendations focused on the type of force plate, the subject's foot posture, and the choice of assessment variables, which further enriched the content of posturography. The authors concluded that in order to promote a more reasonable balance assessment method based on force plates, further methodological research and a stronger consensus are still needed.

Epley Omniax Repositioning Chair for the Treatment of Benign Paroxysmal Positional Vertigo

West et al (2016) examined the clinical value of repositioning chairs in management of patients with refractory BPPV and investigated how different BPPV subtypes would respond

to treatment. In a retrospective study, these researchers performed a chart review of 150 consecutive cases with refractory vertigo referred to their clinic within a 10-month period. Patients with BPPV were managed with classical manual maneuvers, the Epley Omniax Rotator or the TRV chair (TRV). Furthermore, a comprehensive review of the literature was conducted. BPPV was identified in 95 cases. The number of needed treatments for posterior canalolithiasis versus posterior cupulolithiasis, horizontal cupulolithiasis and multi-canal affection was significant ($p < 0.01$); 37 (38 %) patients required only 1 repositioning maneuver and the overall symptom relief was 91.7 % to 100 % after 3 treatments; 11 patients (12 %) experienced relapse within the 6-month follow-up period. Horizontal cupulolithiasis and multi-canal affection constituted the most resilient cases. The literature search identified 9 repositioning chair studies. The Epley Omniax Rotator and the TRV are highly valuable assets in diagnosis and management of BPPV of particularly complex and refractory cases; however, further validation is needed via controlled clinical trials given the high number of BPPV subtypes. These researchers stated that the future will hopefully bring more research on the treatment of BPPV in bi-axial rotational chairs.

Abdulovski and Klokke (2021) stated that multi-axial repositioning chairs such as the TRV chair and the Epley Omniax Rotator are newer alternatives in the treatment of complex and recurrent cases of BPPV. In a systematic review, these investigators examined the available evidence on the clinical characteristics of repositioning chairs for the treatment of BPPV. They carried out a systematic search of the PubMed and Embase databases and data regarding clinical characteristics were extracted from both retrospective and prospective studies, and a qualitative synthesis was made. Of 36 unique publications, 9 studies were considered eligible, containing data from 3,383 subjects; no RCTs were found. The included studies were found to have a high-risk of bias and the overall quality of evidence was low. The type of referred patients and follow-up periods varied. Recurrence rates varied between 11 % and 27.9 %. Incidence of rarer types of BPPV was higher in the included studies than previous estimates. The rate of symptom relief was high, and clinical outcomes were similar between posterior canal BPPV (P-BPPV) and non-P BPPV. The authors concluded that the included studies showed repositioning chairs to be a safe and effective treatment for BPPV, especially for rarer forms and in patients unable to perform manual treatment. Moreover, these researchers noted that the lack of RCTs did not allow for direct comparison of treatment effectiveness with manual CRP. They stated that data from prospective RCTs are needed to compare with conventional methods to examine their effectiveness, to determine indications for treatment, and to decide whether they should be used as 1st-line treatment.

The authors stated that limitations of this review primarily related to the on the low-quality of evidence in the reviewed studies. The studies were either retrospective or prospective without a blinded setup or control group, resulting in a risk of bias. Moreover, 1 of the included retrospective studies treated posterior canalolithiasis with both manual CRP and repositioning chairs, without randomization, which also made the effectiveness of the intervention uncertain. One prospective study had a comparison intervention; however, no description of randomization was reported and the difference between sample sizes for the primary and comparison intervention groups was not explained. These 2 aspects posed a large risk of selection bias. The data in the included studies in this review were heterogeneous and sometimes incomplete, in some cases because the studies had different endpoints.

In an article on “Positioning devices for BPPV treatment” that discussed the Epley Omniax device and the TRV Chair”, Timothy Hain, M.D. (2021) noted that “We are presently not sure whether or not these devices are more effective than conventional methods, and we are also not sure whether or not they pose any unusual risks. With regards to the latter, it would seem to us that prolonged, or upside-down positioning procedures might run some risk of complications including glaucoma, “canal jamming”, and perhaps movement of loose otoconia into the endolymphatic duct. It will take experience to be sure. What has been written so far seems to support their use. There have been a few publications. Searching for “Omniax” in PubMed brings up very little, and it takes sleuthing to find articles that talk about these devices”.

Telemedicine for the Management of Chronic Vertigo

In a systematic review, Meinhardt et al (2023) examined the available evidence on the use of telemedicine to evaluate, diagnose, and manage patients with dizziness. Data sources included Web of Science, SCOPUS, and Medline PubMed databases. The inclusion criteria

included the following: pertaining to telemedicine and the evaluation, diagnosis, treatment, or management of dizziness. Exclusion criteria included the following: single-case studies, meta-analyses, as well as literature and systematic reviews. Outcomes recorded for each article included the following: study type, patient population, telemedicine format, dizziness characteristics, level of evidence, and quality assessment. The search returned 15,408 articles, and a team of 4 screened the articles for inclusion criteria status. A total of 9 studies met the inclusion criteria and were included for review. Of the 9 studies, four were randomized clinical trials, 3 were prospective, cohort studies, and 2 were qualitative studies. The telemedicine format was synchronous in 3 studies and asynchronous in 6 studies; 2 of the studies involved acute dizziness only, 4 involved chronic dizziness only, 1 involved both acute as well as chronic dizziness, and 2 did not specify dizziness type; 6 of the studies included the diagnosis of dizziness, 2 involved the evaluation of dizziness, and 3 involved treatment/management. Some of the reported benefits of telemedicine for dizziness patients included cost-savings, convenience, high patient satisfaction, as well as improvement in dizziness symptoms. The authors concluded that few studies examined the evaluation, diagnosis, or management of dizziness using telemedicine. The lack of protocols and standards of care (SOC) for telemedicine evaluation of dizzy patients creates some challenges in care delivery; however, these reviewed studies provided examples of the breadth of care that has been provided remotely. Moreover, these investigators stated that drawbacks of this systematic review included access to telemedicine technology, Internet connectivity, and dizziness symptoms interfering with the telemedicine application.

GyroStim Machine

The GyroStim was cleared by the FDA via the 510(k) process (No. K220231). It is intended to assist in the treatment of balance disorders and vestibular dysfunction in adult patients, aged 18 years and older, up to 400 lb. It is intended to be used by a trained medical professional as prescribed by physician in a clinical environment.

There is lack of evidence regarding the effectiveness of the GyroStim machine.

References

The above policy is based on the following references:

Benign Paroxysmal Positioning Vertigo

1. Appiani GC, Gagliardi M, Urbani L, Lucertini M. The Epley maneuver for the treatment of benign paroxysmal positional vertigo. *Eur Arch Otorhinolaryngol*. 1996;253(1-2):31-34.
2. Barton JJS. Benign paroxysmal positional vertigo. UpToDate [online serial]. Waltham, MA: UpToDate; reviewed December 2020.
3. Black FO, Angel CR, Pesznecker SC, et al. Outcome analysis of individualized vestibular rehabilitation protocols. *Am J Otol*. 2000;21(4):543-551.
4. Brandt T, Stedden S, Daroff RB. Therapy for benign paroxysmal positional vertigo, revisited. *Neurology*. 1994;44(5):796-800.
5. Bromwich M, Hughes B, Raymond M, et al. Efficacy of a new home treatment device for benign paroxysmal positional vertigo. *Arch Otolaryngol Head Neck Surg*. 2010;136(7):682-685.
6. Bromwich MA, Parnes LS. The DizzyFix: Initial results of a new dynamic visual device for the home treatment of benign paroxysmal positional vertigo. *J Otolaryngol Head Neck Surg*. 2008;37(3):380-387.
7. Brooks JG, Abidin MR. Repositioning maneuver for benign paroxysmal vertigo (BPPV). *J Am Osteopath Assoc*. 1997;97(5):277-279.
8. Buintjes TD, Companjen J, van der Zaag-Loonen HJ, van Benthem PP. A randomised sham-controlled trial to assess the long-term effect of the Epley manoeuvre for treatment of posterior canal BPPV. *Clin Otolaryngol*. 2014;39(1):39-44.
9. Dornhoffer JL, Colvin GB. Benign paroxysmal positional vertigo and canalith repositioning clinical correlations. *Am J Otol*. 2000;21(2):230-233.

10. El-Kashlan HK, Shepard NT, Asher AM, et al. Evaluation of clinical measures of equilibrium. *Laryngoscope*. 1998;108(3):311-319.
11. Epley JM. The canalith repositioning procedure: For treatment of benign paroxysmal positional vertigo. *Otolaryngol Head Neck Surg*. 1992;107:399-404.
12. Fife TD, Iverson DJ, Lempert T, et al; Quality Standards Subcommittee, American Academy of Neurology. Practice parameter: Therapies for benign paroxysmal positional vertigo (an evidence-based review): Report of the Quality Standards Subcommittee of the American Academy of Neurology. *Neurology*. 2008;70(22):2067-2074.
13. Fung K, Hall SF. Particle repositioning maneuver: Effective treatment for benign paroxysmal positional vertigo. *J Otolaryngol*. 1996;25(4):243-248.
14. Herdman SJ, Tusa RJ, Zee DS, et al. Single treatment approaches to benign paroxysmal positional vertigo. *Arch Otolaryngol Head Neck Surg*. 1993;119(4):450-454.
15. Hilton M, Pinder D. The Epley (canalith repositioning) manoeuvre for benign paroxysmal positional vertigo. *Cochrane Database Syst Rev*. 2004;(2):CD003162.
16. Huebner AC, Lytle SR, Doettl SM, et al. Treatment of objective and subjective benign paroxysmal positional vertigo. *J Am Acad Audiol*. 2013;24(7):600-606.
17. Labuguen RH. Initial evaluation of vertigo. *Am Fam Physician*. 2006;73(2):244-251.
18. Lempert T, Gresty MA, Bronstein AM. Benign positional vertigo: Recognition and treatment. *Br Med J*. 1995;311(7003):489-491.
19. Maas BDPJ, van der Zaag-Loonen HJ, van Benthem PPG, Bruintjes TD. Effectiveness of canal occlusion for intractable posterior canal benign paroxysmal positional vertigo: A systematic review. *Otolaryngol Head Neck Surg*. 2020;162(1):40-49.
20. Nunez RA, Cass SP, Furman JM. Short- and long-term outcomes of canalith repositioning for benign paroxysmal positioning vertigo. *Otolaryngol Head Neck Surg*. 2000;122(5):647-652.
21. Ruckenstein MJ, Shepard NT. Balance function testing: A rational approach. *Otolaryngol Clin North Am*. 2000;33(3):507-518.
22. Semont A, Freyss G, Vitte E. Curing the BPPV with a laboratory maneuver. *Adv Otorhinolaryngol*. 1988;42:290-293.
23. Silva AL, Marinho MR, Gouveia FM, et al. Benign paroxysmal positional vertigo: Comparison of two recent international guidelines. *Braz J Otorhinolaryngol*. 2011;77(2):191-200.
24. White J, Savvides P, Cherian N, Oas J. Canalith repositioning for benign paroxysmal positional vertigo. *Otol Neurotol*. 2005;26(4):704-710.
25. Wolf JS, Boyev KP, Manokay BJ, Mattox DE. Success of the modified Epley maneuver in treating benign paroxysmal positional vertigo. *Laryngoscope*. 1999;109(6):900-903.
26. Zhu Q, Liu C, Lin C, et al. Efficacy and safety of semicircular canal occlusion for intractable horizontal semicircular benign paroxysmal positional vertigo. *Ann Otol Rhinol Laryngol*. 2015;124(4):257-260.

Vestibular Rehabilitation

1. Burton MJ, Monsell EM, Rosenfeld RM. Extracts from The Cochrane Library: Vestibular rehabilitation for unilateral peripheral vestibular dysfunction (review). *Otolaryngol Head Neck Surg*. 2008;138(4):415-417.
2. Cohen H. Vestibular rehabilitation reduces functional disability. *Otolaryngol Head Neck Surg*. 1992;107:638-643.
3. Cowland L, Martin J. Efficacy of vestibular rehabilitation. *Otolaryngol Head Neck Surg*. 1998;1:49-54.
4. Dunlap PM, Holmberg JM, Whitney SL. Vestibular rehabilitation: Advances in peripheral and central vestibular disorders. *Curr Opin Neurol*. 2019;32(1):137-144.
5. El-Kashlan HK, Shepard NT, Arts HA, et al. Disability from vestibular symptoms after acoustic neuroma resection. *Am J Otol*. 1998;19(1):104-111.
6. Eldoen G, Ljostad U, Goplen FK, et al. Persistent postural-perceptual dizziness. *Tidsskr*

Nor Laegeforen. 2019;139(9).

7. Furman JM, Barton JJS. Treatment of vertigo. UpToDate [online serial]. Waltham, MA: UpToDate; reviewed January 2022.
8. Girardi M, Konrad HR. Vestibular rehabilitation therapy for the patient with dizziness and balance disorders. *ORL-Head Neck Nursing*. 1998;16(4):13-22.
9. Herdman SJ, Clendaniel RA, Mattox DE, et al. Vestibular adaptation exercises and recovery: Acute stage after acoustic neuroma resection. *Otolaryngol Head Neck Surg*. 1995;113(1):77-87.
10. Herdman SJ. Vestibular rehabilitation. *Curr Opin Neurol*. 2013;26(1):96-101.
11. Hillier SL, McDonnell M. Vestibular rehabilitation for unilateral peripheral vestibular dysfunction. *Cochrane Database Syst Rev*. 2011;(2):CD005397.
12. Horak FB, Jones-Rycewicz C, Black FO, Shumway-Cook A. Effects of vestibular rehabilitation on dizziness and imbalance. *Otolaryngol Head Neck Surg*. 1992;106(2):175-180.
13. Medeiros IR, Bittar RS, Pedalini ME, et al. Vestibular rehabilitation therapy in children. *Otol Neurotol*. 2005;26(4):699-703.
14. Meinhardt G, Perez N, Sharrer C. The role of telemedicine for evaluation and management of dizzy patients: A systematic review. *Otol Neurotol*. 2023;44(5):411-417.
15. Meldrum D, Herdman S, Vance R, et al. Effectiveness of conventional versus virtual reality-based balance exercises in vestibular rehabilitation for unilateral peripheral vestibular loss: Results of a randomized controlled trial. *Arch Phys Med Rehabil*. 2015;96(7):1319-1328.
16. Mempo E, Lau K, Green F, et al. Customised vestibular rehabilitation with the addition of virtual reality based therapy in the management of persistent postural-perceptual dizziness. *J Laryngol Otol*. 2021;135(10):887-891.
17. Mruzek M, Barin K, Nichols DS, et al. Effects of vestibular rehabilitation and social reinforcement on recovery following ablative vestibular surgery. *Laryngoscope*. 1995;105(7 Pt 1):686-692.
18. Strupp M, Arbusow V, Maag KP, et al. Vestibular exercises improve central vestibulospinal compensation after vestibular neuritis. *Neurology*. 1998;51(3):838-844.
19. Trinidad A, Goebel JA. Persistent postural-perceptual dizziness-A systematic review of the literature for the balance specialist. *Otol Neurotol*. 2018;39(10):1291-1303.
20. Vestibular Disorders Association (VeDA). Persistent postural-perceptual dizziness. Portland, OR: VeDA; 2021. Available at: <https://vestibular.org/article/diagnosis-treatment/types-of-vestibular-disorders/persistent-postural-perceptual-dizziness/>. Accessed February 22, 2022.
21. Whitley SL, Rossi MM. Efficacy of vestibular rehabilitation. *Otolaryngol Clin North Am*. 2000;33(3):659-672.
22. Yardley L, Burgneay J, Andersson G, et al. Feasibility and effectiveness of providing vestibular rehabilitation for dizzy patients in the community. *Clin Otolaryngol*. 1998;23(5):442-448.

Dynamic Posturography

1. Alahmari KA, Marchetti GF, Sparto PJ, et al. Estimating postural control with the balance rehabilitation unit: Measurement consistency, accuracy, validity, and comparison with dynamic posturography. *Arch Phys Med Rehabil*. 2014;95(1):65-73.
2. American Academy of Neurology. Assessment: Posturography. Report of the Therapeutics and Technology Assessment Subcommittee of the American Academy of Neurology. *Neurology*. 1993;43(6):1261-1264.
3. American Medical Association (AMA). Computerized dynamic posturography (Equitest). Tech Brief. Chicago, IL: AMA; February 1994:1-5.
4. Baloh RW, Jacobson KM, Beykirch K, Honrubia V. Static and dynamic posturography in

patients with vestibular and cerebellar lesions. *Arch Neurol*. 1998;55(5):649-654.

5. Baloh RW, Jacobson KM, Enrietto JA, et al. Balance disorders in older persons: Quantification with posturography. *Otolaryngol Head Neck Surg*. 1998;119(1):89-92.
6. Baloh RW, Spain S, Socotch TM, et al. Posturography and balance problems in older people. *J Am Geriatr Soc*. 1995;43(6):638-644.
7. Bergson E, Sataloff RT. Preoperative computerized dynamic posturography as a prognostic indicator of balance function in patients with acoustic neuroma. *Ear Nose Throat J*. 2005;84(3):154-156.
8. Dewan BM, James CR, Kumar NA, Sawyer SF. Kinematic validation of postural sway measured by Biodex Biosway (force plate) and SWAY Balance (accelerometer) technology. *Biomed Res Int*. 2019;2019:8185710.
9. Di Fabio RP. Sensitivity and specificity of platform posturography for identifying patients with vestibular dysfunction. *Phys Ther*. 1995;75(4):290-305.
10. Di Girolamo S, Picciotti P, Sergi B, et al. Postural control and glycerol test in Meniere's disease. *Acta Otolaryngol*. 2001;121(7):813-817.
11. Dobie RA. Does computerized dynamic posturography help us care for our patients? *Am J Otol*. 1997;18:108-112.
12. Evans MK, Krebs DC. Posturography does not test vestibulospinal function. *Otolaryngol Head Neck Surg*. 1999;120(2):164-173.
13. Furman JM. Role of posturography in the management of vestibular patients. *Otolaryngol Head Neck Surg*. 1995;112(1):8-15.
14. Harstall, C. Dynamic posturography in the rehabilitation of stroke, brain injured and amputee patients. HTA 7. Edmonton, AB: Alberta Heritage Foundation for Medical Research (AHFMR); February 1998.
15. Karartı C, Ozudogru A, Basat HC, et al. Determination of Biodex Balance System cutoff scores in older people with nonspecific back pain: A cross-sectional study. *J Manipulative Physiol Ther*. 2021;44(1):85-94.
16. Kohle F, Stark C, Klunter H-D, et al. Peripheral neuropathy, an independent risk factor for falls in the elderly, impairs stepping as a postural control mechanism: A case-cohort study. *J Peripher Nerv Syst*. 2024;29(4):453-463.
17. Miner DG, Harper BA, Glass SM. Validity of postural sway assessment on the Biodex BioSway™ compared with the NeuroCom Smart Equitest. *J Sport Rehabil*. 2020;30(3):516-520.
18. O'Neill DE, Gill-Body KM, Krebs DE. Posturography changes do not predict functional performance changes. *Am J Otol*. 1998;19(6):797-803.
19. Parsa M, Rahimi A, Dehkordi SN. Studying the correlation between balance assessment by Biodex Stability System and Berg Scale in stroke individuals. *J Bodyw Mov Ther*. 2019;23(4):850-854.
20. Piirtola M, Era P. Force platform measurements as predictors of falls among older people - a review. *Gerontology*. 2006;52(1):1-16.
21. Ruckenstein MJ, Shepard NT. Balance function testing: A rational approach. *Otolaryngol Clin North Am*. 2000;33(3):507-518.
22. Rudge P, Bronstein AM. Investigations of disorders of balance. *J Neurol Neurosurg Psychiatr*. 1995;59:568-578.
23. Soto A, Labella T, Santos S, et al. The usefulness of computerized dynamic posturography for the study of equilibrium in patients with Meniere's disease: Correlation with clinical and audiologic data. *Hear Res*. 2004;196(1-2):26-32.
24. Stella I, Perrin P, Casteran M, et al. Utilization of vestibular information for balance control in children with Chiari I malformation. *Audiol Res*. 2024;14(6):939-957.
25. U.S. Department of Health and Human Services, Health Care Financing Administration (HCFA). Computerized dynamic posturography. Technology Advisory Committee Minutes, May 6 - 7, 1997. Baltimore, MD; HCFA; 1997:2-7.

Meniett Device

1. American Academy of Otolaryngology-Head and Neck Surgery (AAO-HNS). AAO-HNS Position on Micropressure Therapy. Alexandria, VA: AAO-HNS; March 2008. Available at: <http://www.entnet.org/Practice/micropressure.cfm>. Accessed February 11, 2009.
2. Barbara M, Consagra C, Monini S, et al. Local pressure protocol, including Meniett, in the treatment of Meniere's disease: Short-term results during the active stage. *Acta Otolaryngol*. 2001;121(8):939-944.
3. Barbara M, Monini S, Chiappini I, Filippo R. Meniett therapy may avoid vestibular neurectomy in disabling Meniere's disease. *Acta Otolaryngol*. 2007;127(11):1136-1141.
4. Boudewyns AN, Wuyts FL, Hoppenbrouwers M, et al. Meniett therapy: Rescue treatment in severe drug-resistant Meniere's disease. *Acta Otolaryngol* 2005;125(12):1283-1289.
5. da Costa SS, de Sousa LC, Piza MR. Meniere's disease: Overview, epidemiology, and natural history. *Otolaryngol Clin North Am*. 2002;35(3):455-495.
6. Dornhoffer JL, King D. The effect of the Meniett device in patients with Ménière's disease: Long-term results. *Otol Neurotol*. 2008;29(6):868-874.
7. Gates GA, Green JD Jr, Tucci DL, Telian SA. The effects of transtympanic micropressure treatment in people with unilateral Meniere's disease. *Arch Otolaryngol Head Neck Surg*. 2004;130(6):718-725.
8. Gates GA, Green JD Jr. Intermittent pressure therapy of intractable Meniere's disease using the Meniett device: A preliminary report. *Laryngoscope*. 2002;112(8 Pt 1):1489-1493.
9. Gates GA, Verrall A, Green JD Jr, et al. Meniett clinical trial: Long-term follow-up. *Arch Otolaryngol Head Neck Surg*. 2006;132(12):1311-1316.
10. Huang W, Liu F, Gao B, Zhou J. Clinical long-term effects of Meniett pulse generator for Meniere's disease. *Acta Otolaryngol*. 2009;129(8):819-825.
11. Mattox DE, Reichert M. Meniett device for Meniere's disease: Use and compliance at 3 to 5 years. *Otol Neurotol*. 2008;29(1):29-32.
12. National Horizon Scanning Centre (NHSC). Meniett low-pressure pulse generator for Meniere's disease - horizon scanning review. Birmingham, UK: NHSC; 2003.
13. Odkvist LM, Arlinger S, Billermark E, et al. Effects of middle ear pressure changes on clinical symptoms in patients with Meniere's disease -- a clinical multicentre placebo-controlled study. *Acta Otolaryngol Suppl*. 2000;543:99-101.
14. Peterson WM, Isaacson JE. Current management of Ménière's disease in an only hearing ear. *Otol Neurotol*. 2007;28(5):696-699.
15. Rajan GP, Din S, Atlas MD. Long-term effects of the Meniett device in Meniere's disease: The Western Australian experience. *J Laryngol Otol*. 2005;119(5):391-395.
16. Reddy SS, Newlands SD. Treatment controversies in Meniere's disease. *UTMB Otolaryngology Grand Rounds*. Galveston, TX: University of Texas Medical Branch at Galveston; May 18, 2005. Available at: <http://www.utmb.edu/otoref/grnds/Menieres-050518/Menieres-slides-050518.pdf>. Accessed April 5, 2006.
17. Syed MI, Rutka J, Hendry J, Browning GG. Positive pressure therapy for Meniere's syndrome/ disease with a Meniett device: A systematic review of randomised controlled trials. *Clin Otolaryngol*. 2015;40(3):197-207.
18. Thai-Van H, Bounaix MJ, Frayssé B. Meniere's disease: Pathophysiology and treatment. *Drugs*. 2001;61(8):1089-1102.
19. Thomsen J, Sass, K, Odkvist, L, Arlinger S. Local overpressure treatment reduces vestibular symptoms in patients with Meniere's disease: A clinical, randomized, multicenter, double-blind, placebo-controlled study. *Otol Neurotol*. 2005;26(1):68-73.
20. van Sonsbeek S, Pullens B, van Benthem PP. Positive pressure therapy for Meniere's disease or syndrome. *Cochrane Database Syst Rev*. 2015;3:CD008419.

Electronystagmography and Videonystagmography

1. American Academy of Neurology. Assessment: Electronystagmography. Report of the Therapeutics and Technology Assessment Subcommittee. *Neurology*. 1996;46(6):1763-1766.
2. Eggers SD, Zee DS. Evaluating the dizzy patient: Bedside examination and laboratory assessment of the vestibular system. *Semin Neurol*. 2003;23(1):47-58.
3. Fife TD, Tusa RJ, Furman JM, et al. Assessment: Vestibular testing techniques in adults and children: Report of the Therapeutics and Technology Assessment Subcommittee of the American Academy of Neurology. *Neurology*. 2000;55(10):1431-1441.
4. Ganança MM, Caovilla HH, Ganança FF. Electronystagmography versus videonystagmography. *Braz J Otorhinolaryngol*. 2010;76(3):399-403.
5. Markley BA. Introduction to electronystagmography for END technologists. *Am J Electroneurodiagnostic Technol*. 2007;47(3):178-189.
6. Perez N. Vibration induced nystagmus in normal subjects and in patients with dizziness. A videonystagmography study. *Rev Laryngol Otol Rhinol (Bord)*. 2003;124(2):85-90.
7. Petrova D, Hannig A. Electronystagmographic and caloric investigation data about vascular-vestibular dysfunction among patients with vertebrobasilar insufficiency. *Int Tinnitus J*. 2003;9(1):48-51.
8. Shoup AG and Townsley AL. Electronsagmography. *eMedicine Otolaryngology*. Topic 373. Omaha, NE: eMedicine.com; updated January 14, 2008. Available at: <http://www.emedicine.com/ent/topic373.htm>. Accessed August 27, 2008.
9. Worden BF, Blevins NH. Pediatric vestibulopathy and pseudovestibulopathy: Differential diagnosis and management. *Curr Opin Otolaryngol Head Neck Surg*. 2007;15(5):304-309.

Cochlear Hydrops Analysis Masking Procedure (CHAMP) Testing for Meniere's Disease

1. Dinces EA. Meniere disease. UpToDate [online serial]. Waltham, MA: UpToDate; reviewed November 2014.
2. Hong SK, Nam SW, Lee HJ, et al. Clinical observation on acute low-frequency hearing loss without vertigo: The role of cochlear hydrops analysis masking procedure as initial prognostic parameter. *Ear Hear*. 2013;34(2):229-235.
3. Kim BJ, Jung SD, Lee HJ, et al. Effect of hearing loss, age, and gender on the outcome of the cochlear hydrops analysis masking procedure. *Otol Neurotol*. 2015;36(3):472-475.

Brainstem Auditory Evoked Potentials for Evaluation of Vertigo

1. Furman JM, Barton JJS. Evaluation of the patient with vertigo. UpToDate [online serial]. Waltham, MA: UpToDate; reviewed December 2018.
2. He JW, Gong Q, Wang XF, Xiao Z. High stimulus rate brainstem auditory evoked potential in benign paroxysmal positional vertigo. *Eur Arch Otorhinolaryngol*. 2015;272(9):2095-2100.
3. Ji W, Zhang X. Relationship of the changes of cervical MRI, TCD and BAEP in patients with "isolated" vertigo. *Int J Clin Exp Pathol*. 2014;7(8):5171-5176.

Cervical and Ocular Vestibular Evoked Myogenic Potentials

1. Fife TD, Colebatch JG, Kerber KA, et al. Practice guideline: Cervical and ocular vestibular evoked myogenic potential testing: Report of the Guideline Development, Dissemination, and Implementation Subcommittee of the American Academy of Neurology. *Neurology*. 2017;89(22):2288-2296.
2. Gunes A, Karali E, Ural A, Ruzgar F. Evaluation of cervical vestibular evoked myogenic potential measures using different stimulus types in patients with posterior canal benign paroxysmal positional vertigo. *Acta Otolaryngol*. 2020;140(5):395-400.
3. Lee S-U, Kim H-J, Choi J-Y, et al. Abnormal cervical vestibular-evoked myogenic potentials predict evolution of isolated recurrent vertigo into Meniere's disease. *Front Neurol*. 2017;8:463.
4. Semmanaselvan K, Vignesh SS, Muthukumar R, Jaya V. Vestibular evoked myogenic potentials after Epleys manoeuvre among individuals with benign paroxysmal positional vertigo. *Indian J Otolaryngol Head Neck Surg*. 2019;71(2):195-200.

5. Xu M, Chen ZC, Wei XY, et al. Evaluation of vestibular evoked myogenic potential, caloric test and cochlear electrogram in the diagnosis of Meniere's disease. *Lin Chung Er Bi Yan Hou Tou Jing Wai Ke Za Zhi*. 2019;33(8):704-708.

Balance Tracking System (BTrackS) for Evaluation of Risk of Falling

1. Cappleman AS, Thiamwong L. Fear of falling assessment and interventions in community-dwelling older adults: A mixed methods case-series. *Clin Gerontol*. 2020;43(4):471-482.
2. Chen B, Liu P, Xiao F, et al. Review of the upright balance assessment based on the force plate. *Int J Environ Res Public Health*. 2021;18(5):2696.
3. Goble DJ, Baweja HS. Postural sway normative data across the adult lifespan: Results from 6280 individuals on the Balance Tracking System balance test. *Geriatr Gerontol Int*. 2018;18(8):1225-1229.
4. Goble DJ, Hearn MC, Baweja HS. Combination of BTrackS and Geri-Fit as a targeted approach for assessing and reducing the postural sway of older adults with high fall risk. *Clin Interv Aging*. 2017;12:351-357.
5. Goble DJ, Conner NO, Nolff MR, et al. Test-retest reliability of the Balance Tracking System modified clinical test of sensory integration and balance protocol across multiple time durations. *Med Devices (Auckl)*. 2021;14:355-361.
6. Haworth J, Goble D, Pile M, Kendall B. BTrackS limits of stability test is a reliable assessment of volitional dynamic postural control. *Gait Posture*. 2020;80:298-301.
7. Levy SS, Thralls KJ, Kviatkovsky SA. Validity and reliability of a portable balance tracking system, BTrackS, in older adults. *J Geriatr Phys Ther*. 2018;41(2):102-107.

Epley Omniax Repositioning Chair for the Treatment of Benign Paroxysmal Positional Vertigo

1. Abdulovski S, Klokke M. Repositioning chairs in the diagnosis and treatment of benign paroxysmal positional vertigo -- A systematic review. *J Int Adv Otol*. 2021;17(4):353-360.
2. Barton JJS. Benign paroxysmal positional vertigo. UpToDate [online serial]. Waltham, MA: UpToDate; reviewed December 2022.
3. West N, Hansen S, Moller MN, et al. Repositioning chairs in benign paroxysmal positional vertigo: Implications and clinical outcome. *Eur Arch Otorhinolaryngol*. 2016;273(3):573-580.